

Module 5 : Artificial Neural Network Modelling & Application

11 January 2023

Assoc. Prof. Ir. Dr.-Ing. Mohamad Rizza Othman

Trainer Background

- Name : Mohamad Rizza Othman
- Position : Assoc. Prof. FTKKP & Deputy Director PHKK
- Education background
 - B. Eng. (Chem. Eng.) – UTM 2002
 - M. Eng. (Chem. Eng.) – UTM 2005
 - Dr.-Ing – TUB Germany 2011
- Professional qualifications
 - PEng or Ir. (BEM)
 - CEng (Engineering Council UK)
 - Member IChemE UK
 - Grad. Eng./Techn.
 - Aspen Plus Certified User
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Area of Expertise

- >15 years experience in Aspen Plus (Certified User), Dynamics, ASW, EDR and ANN.
- Taught UG/PG courses i.e., Process Control & Dynamics, Process Simulation & Advanced Process Simulation.
- Research area
 - Process modeling and simulation of various processes i.e., biodiesel, acetic acid, hydrogen, oleochemical, distillation, extraction, DWC, MSR, CO₂ absorption etc.
 - Safety analysis and sustainability
 - ANN
 - Optimization and multi criteria decision making
- Consultation projects
 1. Bio-MEG purification with PRSB
 2. Design and development of modular clinical waste incinerator (Radicare Sdn. Bhd.)
 3. Design and development of training rigs (Politeknik Pagoh)

Module Objectives

At the end of this module, you will be able to:

1. Identify the advantages of using ANN/HM and their applications in process industries.
2. Recognize ANN/HM architecture and methodology.
3. Gain the practical skills and knowledge to create ANN/HM.

Outline

0830 – 1030

- Introduction to ANN & its applications
- ANN modelling methodology

1100 – 1300

- ANN model development in MATLAB

1300 – 1400

- Break

1400 – 1700

- Introduction to Hybrid Model (HM)
- HM methodology and its applications
- HM-ANN development in Aspen +

What is AI?

“Artificial Intelligence is the study of how to make computers do things at which, at the moment, people are better.”



Chess game - Kasparov vs Deep Blue



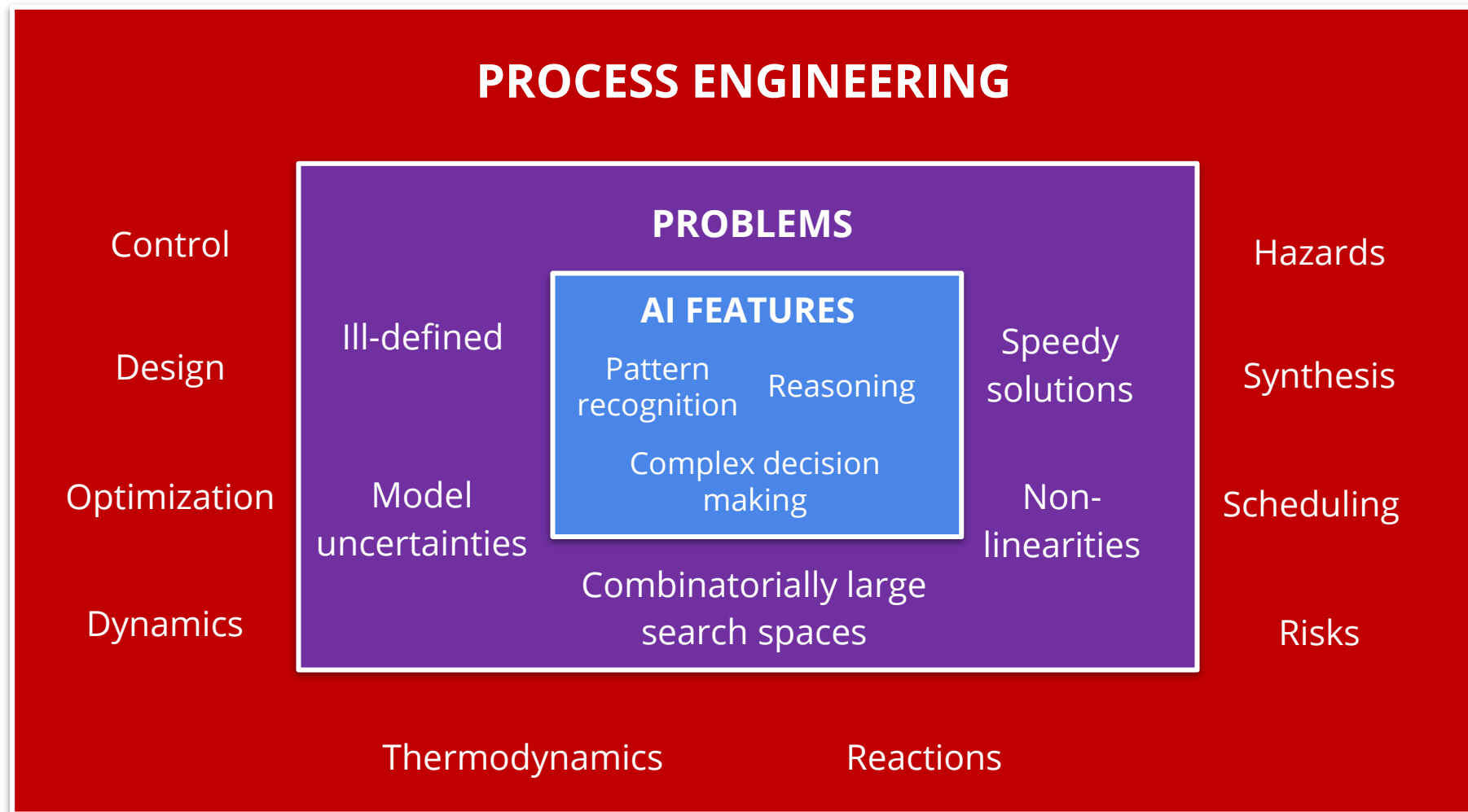
Robotics - Self-driving vehicles etc.



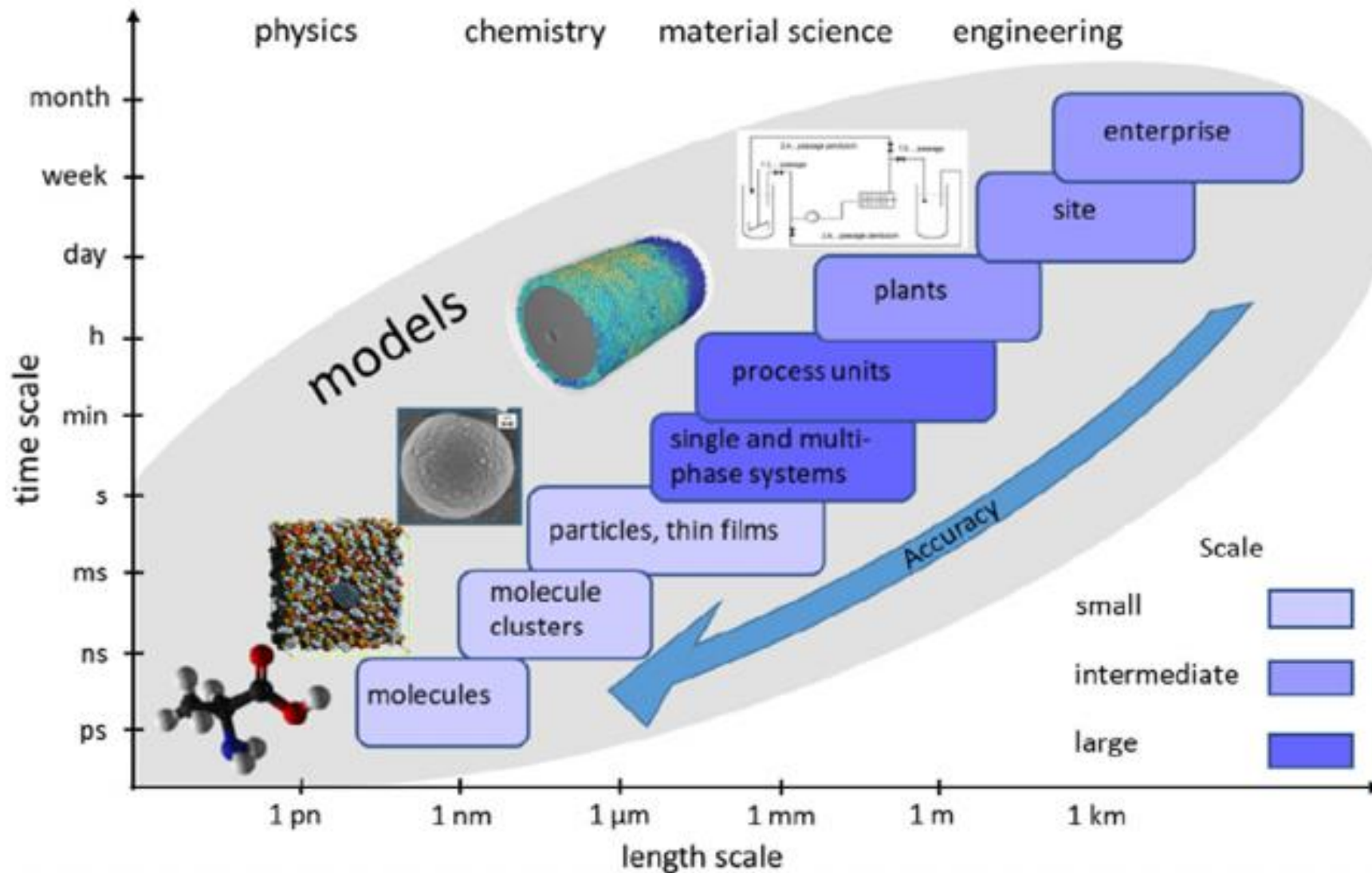
Vision - Self recognition



Speech - Text to speech

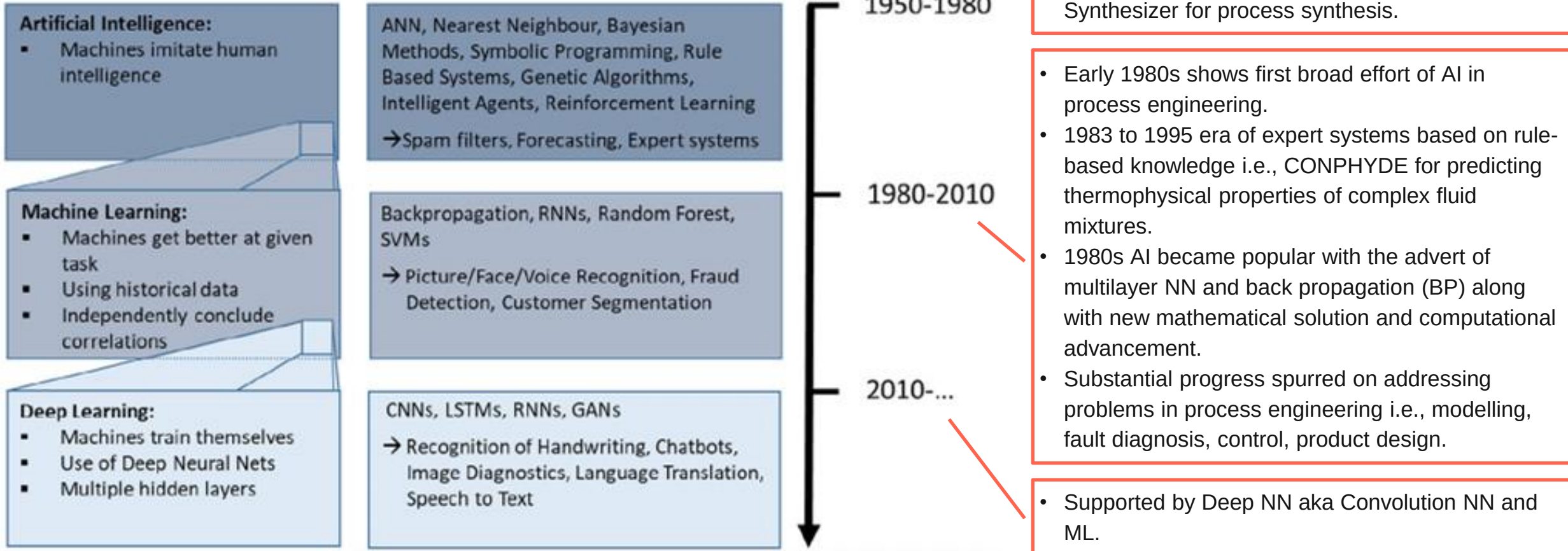


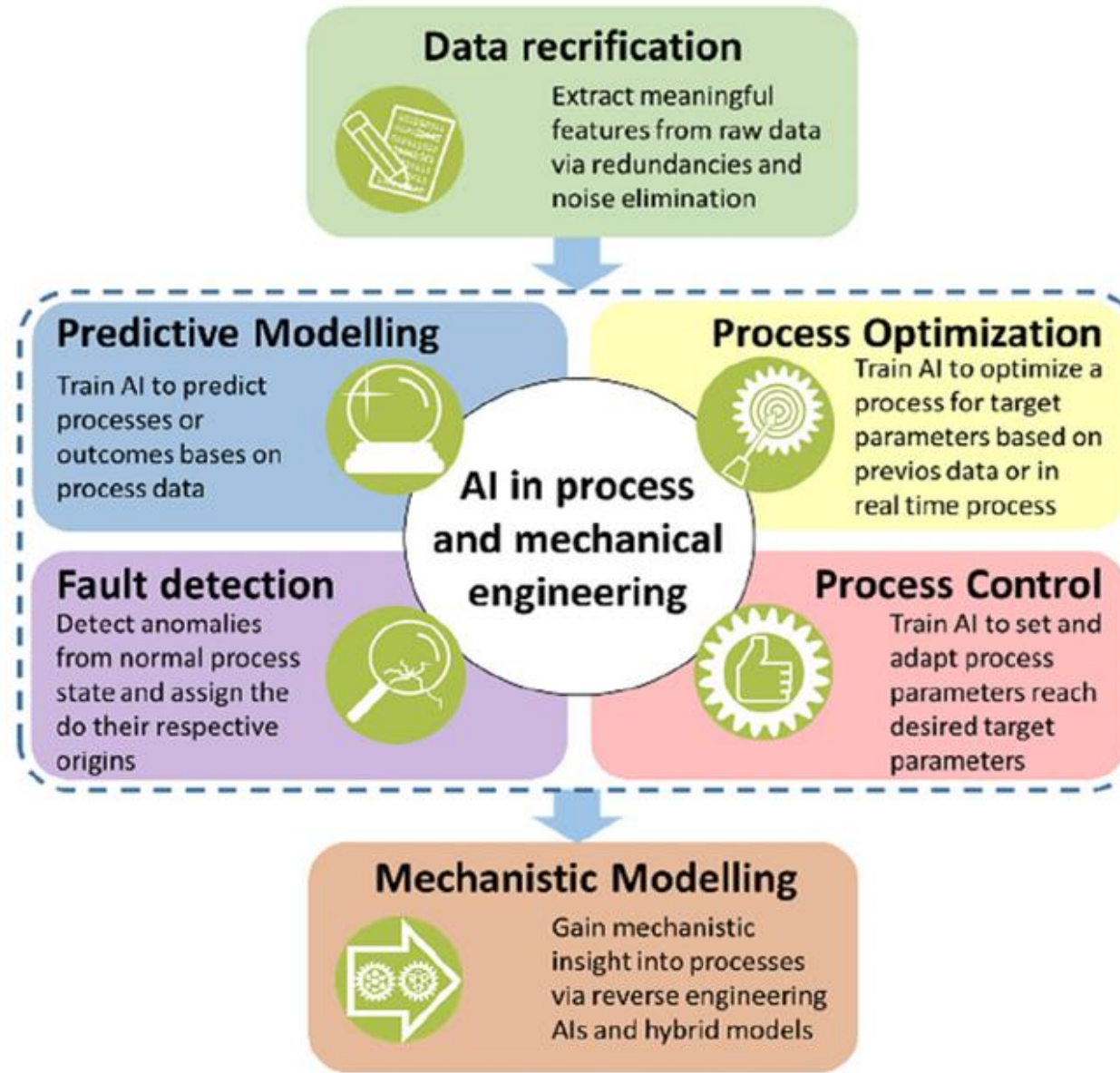
Model spectrum

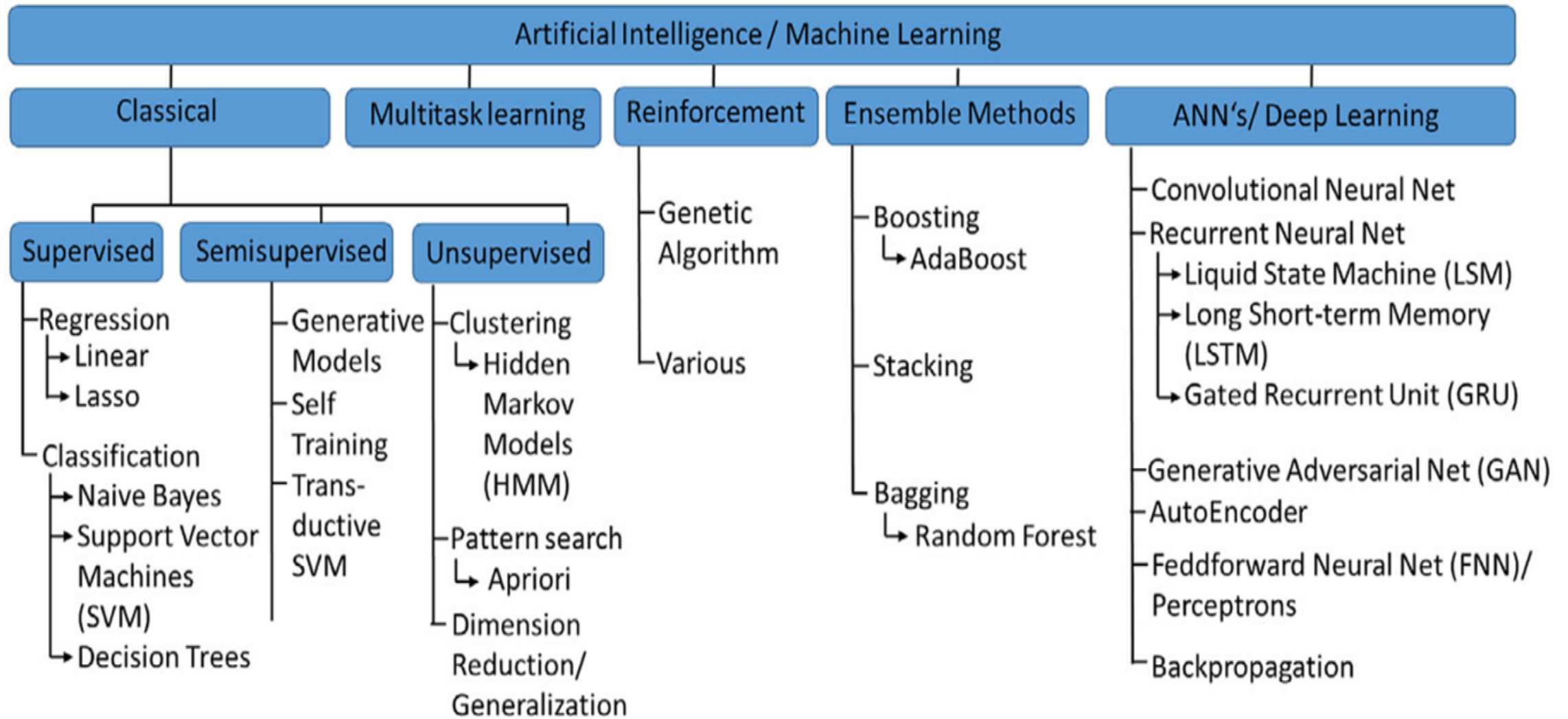




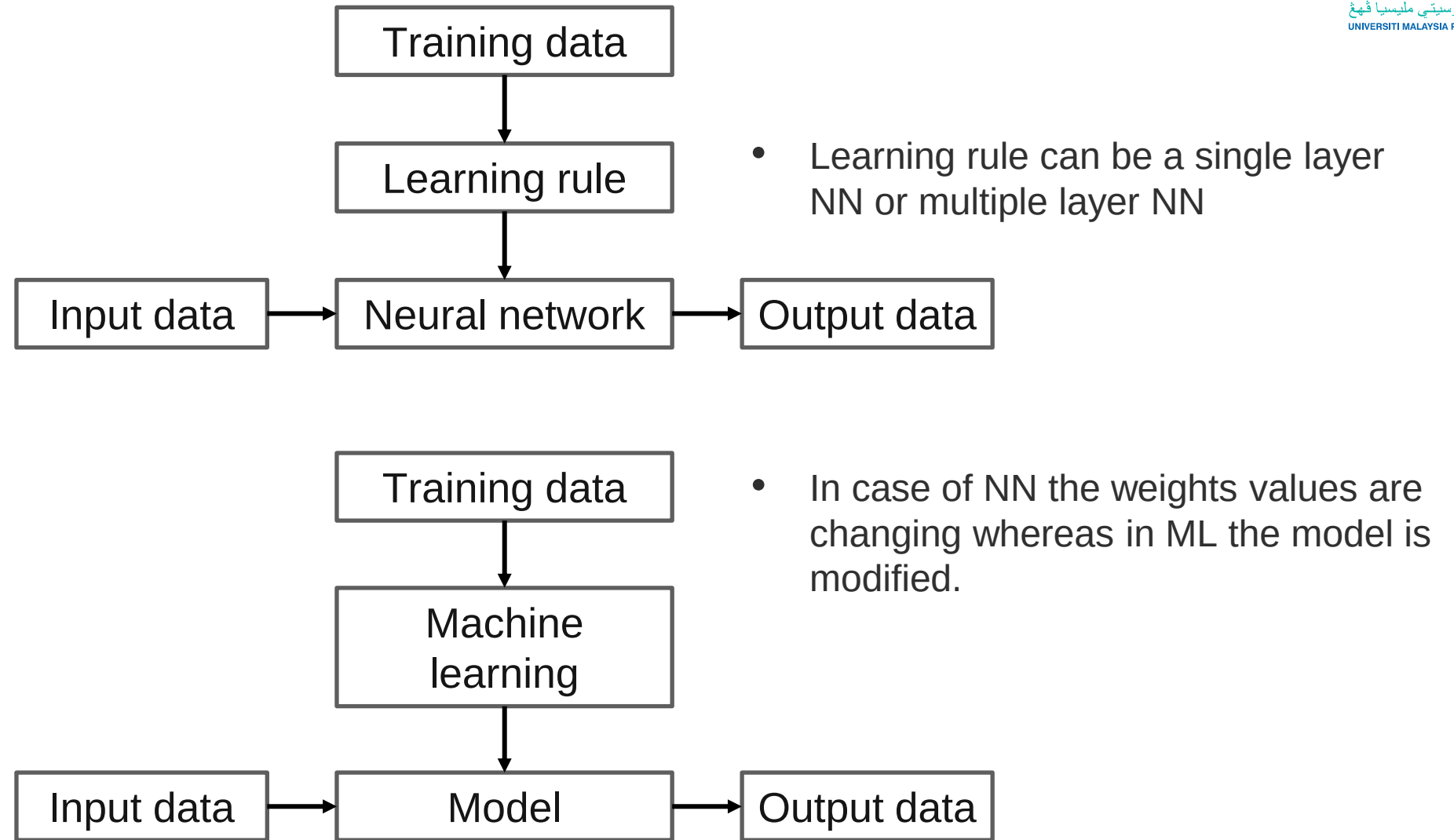
Paradigms of AI & Process Engineering







Taxonomy of AI-Classes and Types.

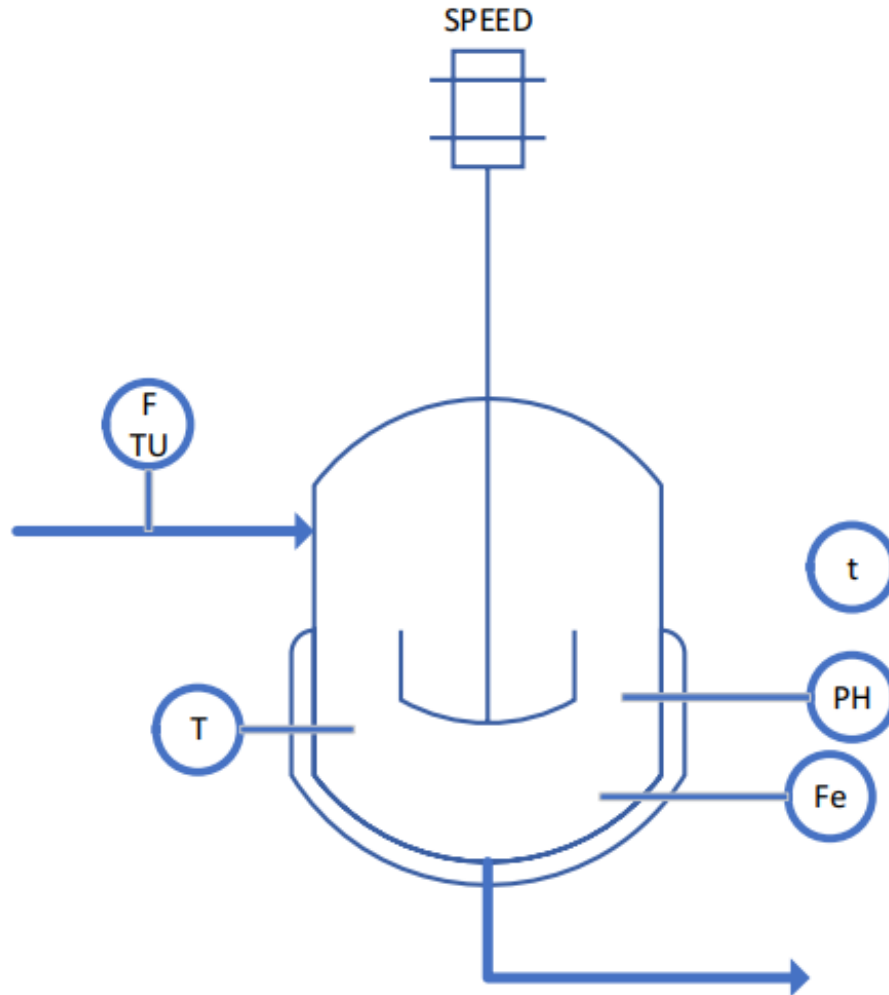


Application

Analysis and Prediction of the Thiourea Gold Leaching Process

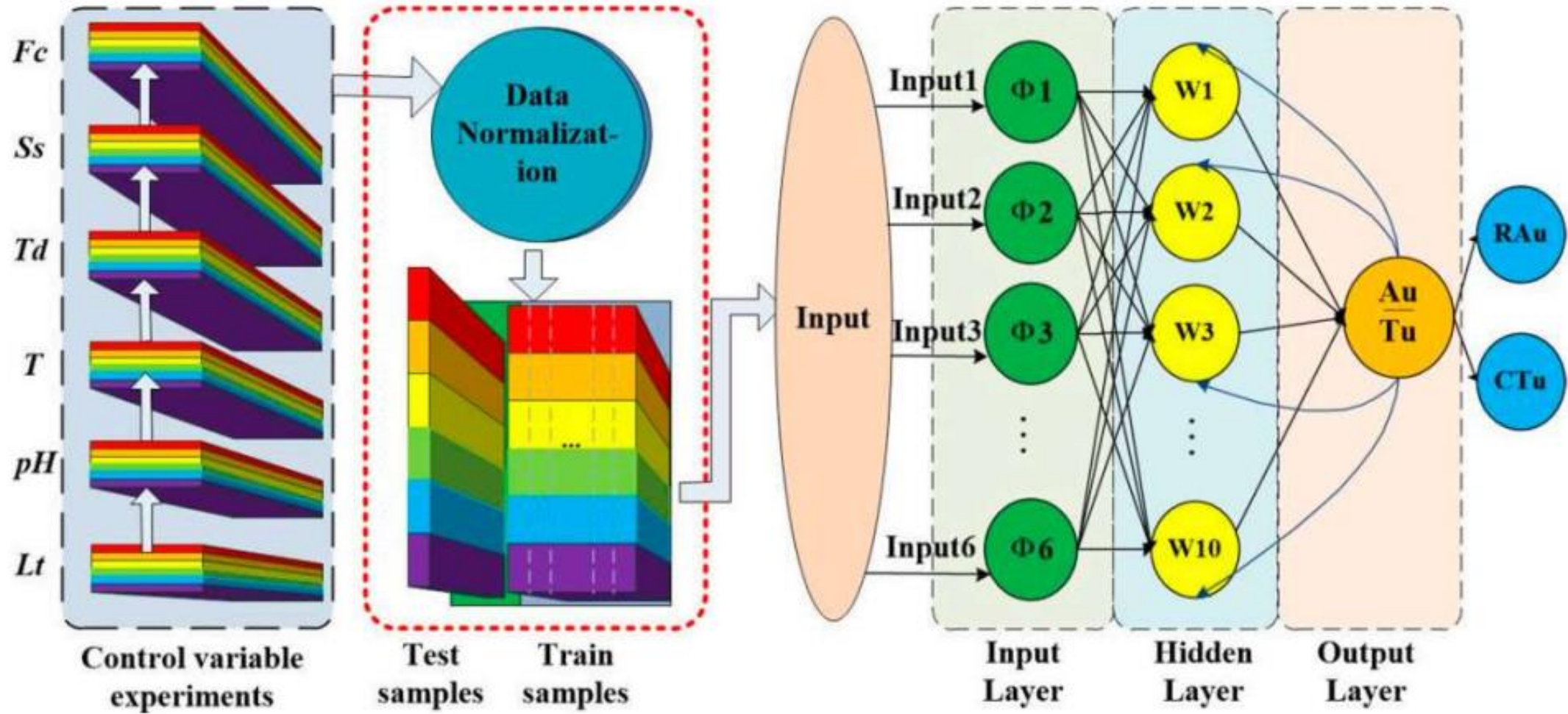
- The thiourea (TU) leaching of gold from refractory ores can be considered an alternative to cyanidation. However, the high reagent consumption causes an increase in cost, which seriously limits its use.
- In order to effectively reduce the TU consumption, it is necessary to analyze the influencing parameters of gold recovery and TU consumption and apply them to the prediction of the TU leaching process.
- Grey relational analysis (GRA) was used to analyze the relational degree between each parameter and gold recovery and TU consumption.
- ANN was established to predict the gold recovery and TU consumption.

Affecting Parameters

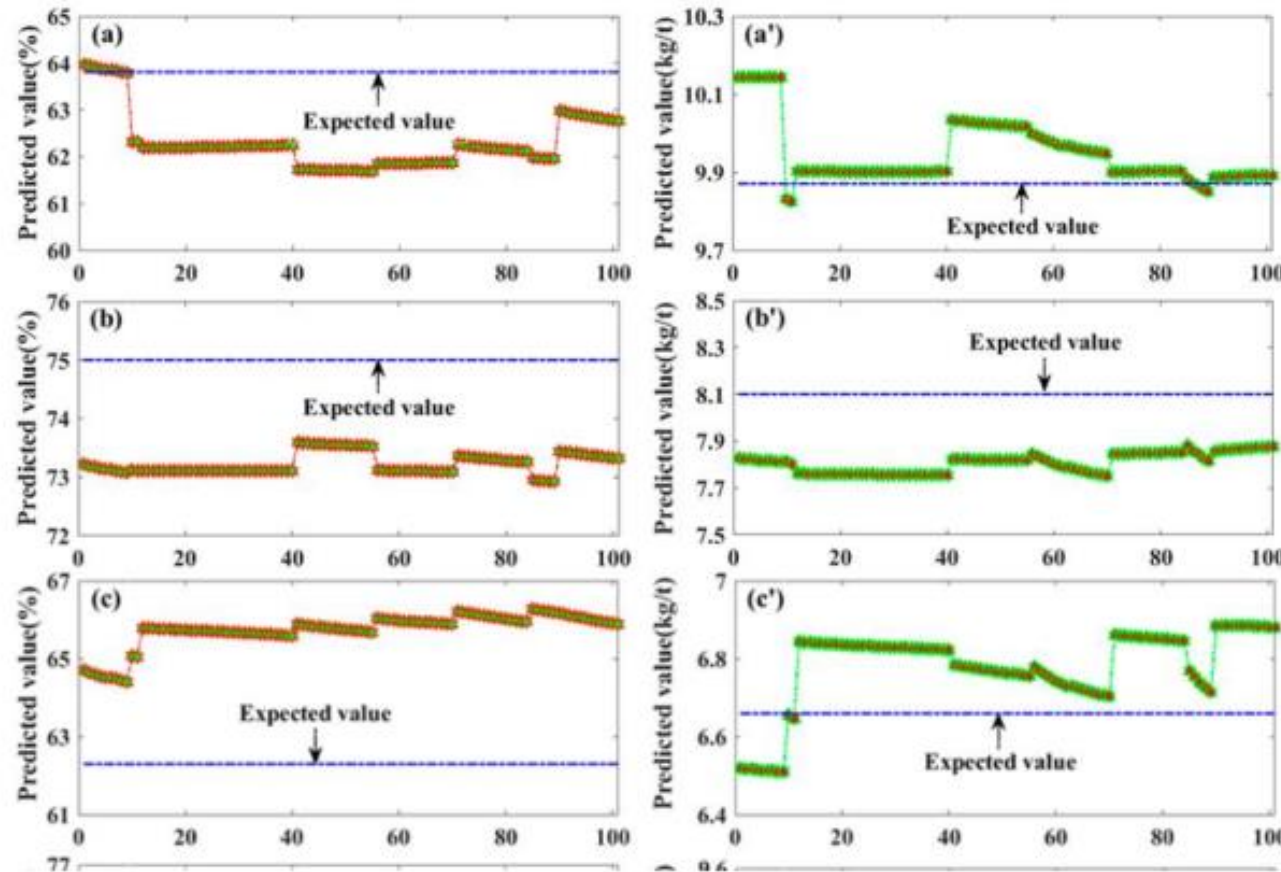


- Parameters related to gold recovery and TU consumption.
 - ❑ t = Leaching time
 - ❑ PH = initial pH
 - ❑ T = temperature
 - ❑ FTU = TU dosage
 - ❑ SPEED = stirring speed
- These parameters will be the input to the ANN.

ANN Structure



Results



- Additional test experiments were used to test the ANN.
- Prediction accuracy of gold recovery is between 94.46% to 98.06%,
- Prediction accuracy for TU consumption is between 95.15% to 99.20%.
- ANN accurately reflect the relationship between the operational conditions and the gold recovery and TU consumption.
- This prediction method can be used as an auxiliary decision-making tool in the TU gold leaching process.

Application

Prediction of Polymer Quality

- Operation and control of industrial polymerization reactors is difficult because of the lack of reliable and timely measurements of key polymer product quality variables close to the reactor.
- Often product samples must be collected hours downstream from the reactor, after the polymer finishing operations.
- Measurement of these quality variables is typically performed off-line in a laboratory and may not be available for several hours after the sample is delivered to the lab.
- Thus, the measurements of the quality variables arrive too late to provide useful feedback for control.

Monomer and polymerization catalyst enters the reactor and mixed. The polymer product, Q_s enters the finishing section to be processed for 2-6 hours. The product quality, Q_{lab} is sampled every 4 hours with which takes another 4 hours to analyze. Online measurement, Q_{on} was unreliable. The reactor was used to manufacture several different polymer grades spanning a wide operating range over which the reactor is highly nonlinear. The polymer finishing section imparts dynamics to the response Q_{lab} to changes in reactor conditions due to mixing.

FF & FT : measure feed flowrate & temperature

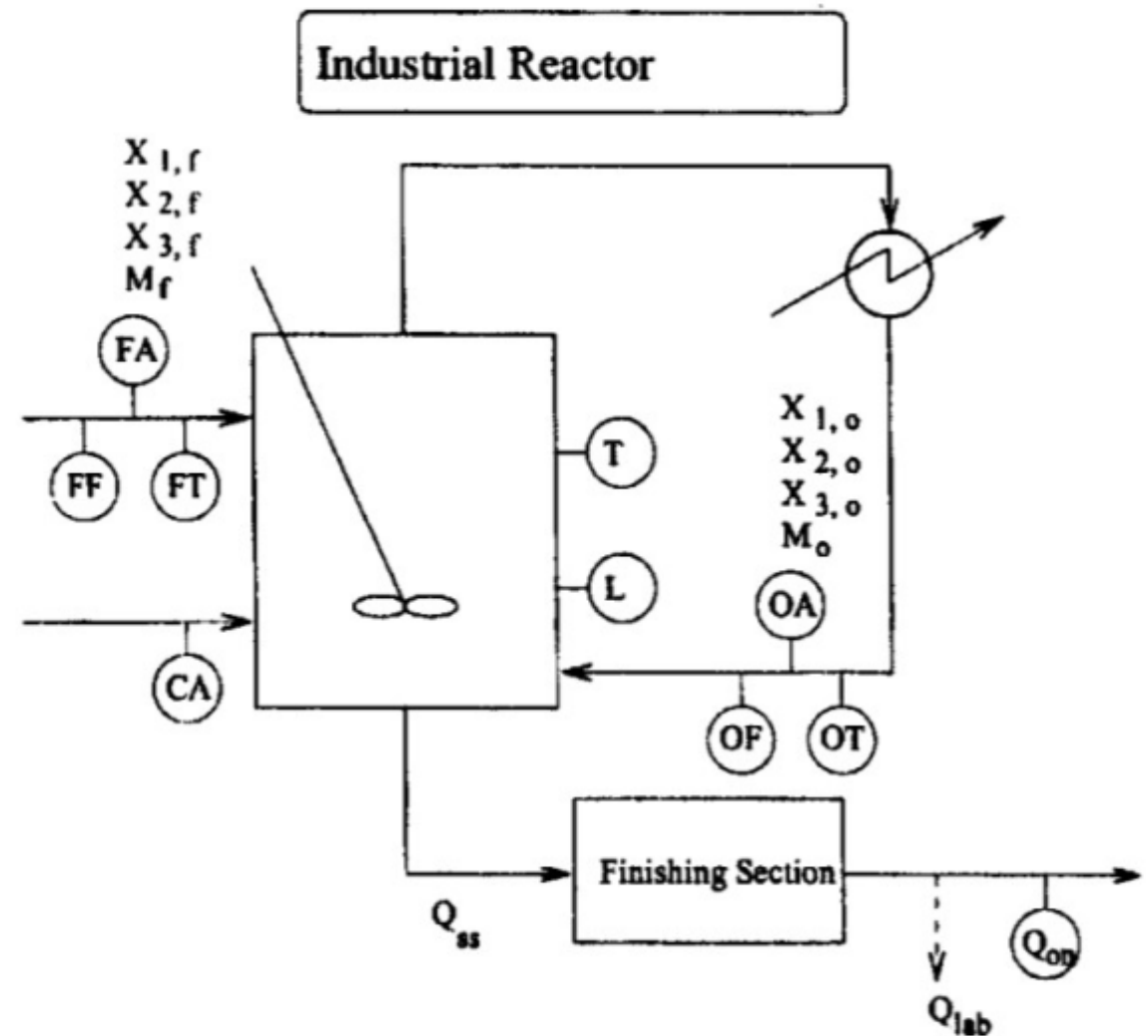
FA : measure monomer concentration, M_f and key impurities, $X_{i,j}$

CA : measure & manipulate polymerization catalyst feed rate

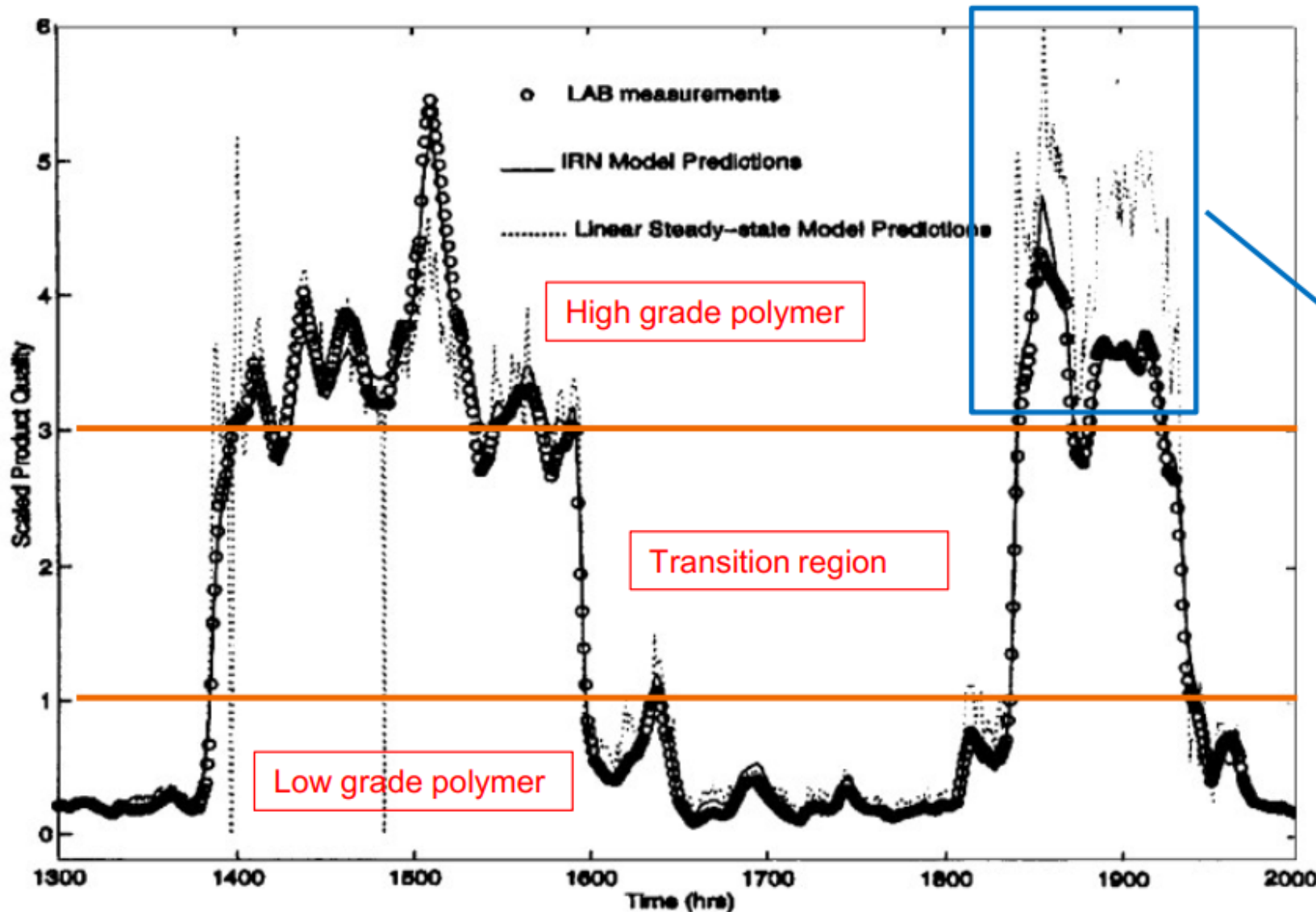
T & L : measure reactor temperature and level

OF & OT : measure recycle flowrate and temperature

OA : measure unreacted monomer, M_o , and key impurities, $X_{i,o}$

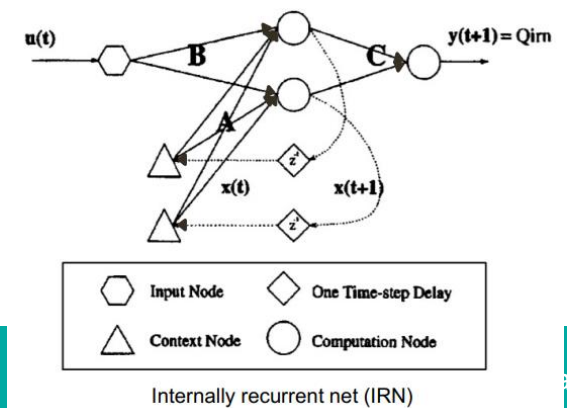


Industrial polymerization reactor and subsequent finishing section (Barton, 1997)



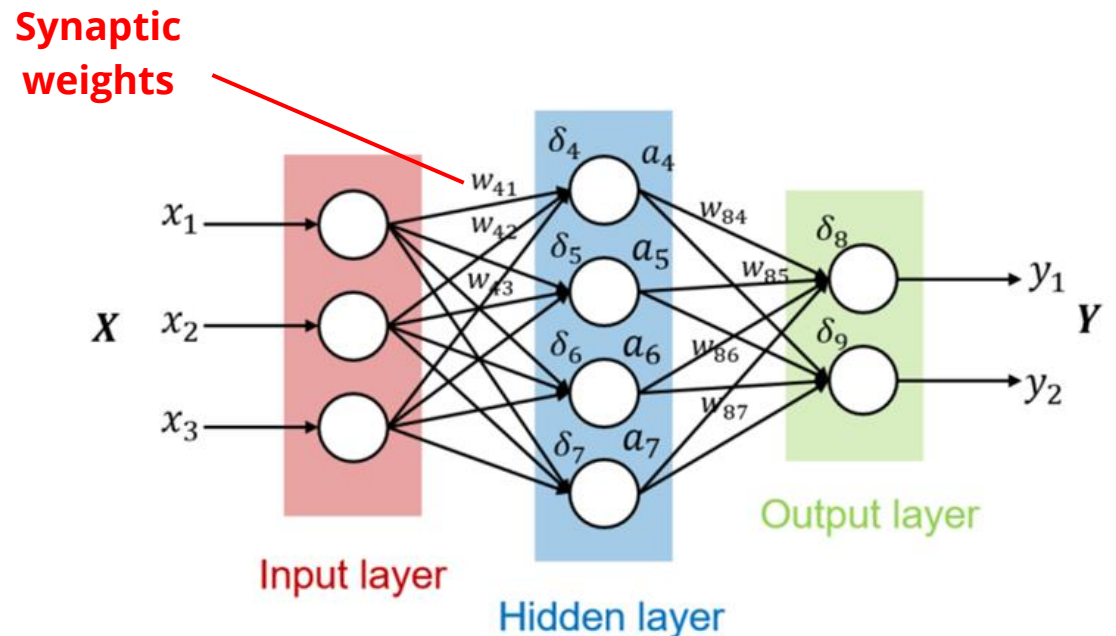
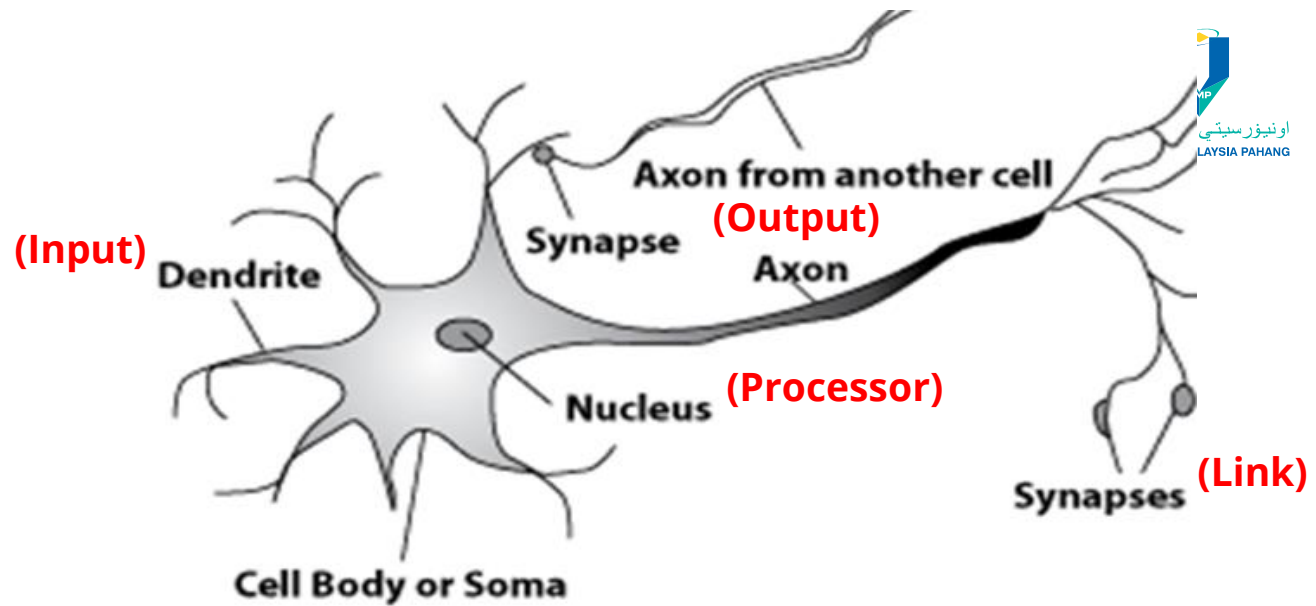
Comparison of polymer product quality predictions for testing data set

- The IRN better able to capture the dynamic characteristics of the reactor and predict product quality several hours before the laboratory measurement become available.
- Show good predictions in all regions.
- Between $t=1800$ to $t=2000$, ANN shows superiority over SS because ANN include the effect of impurity X_2 .
- The IRN model accurately predicts the quality variable over the entire reactor operating region, even between grades, eliminating the need for switching between different models in different operating regions.



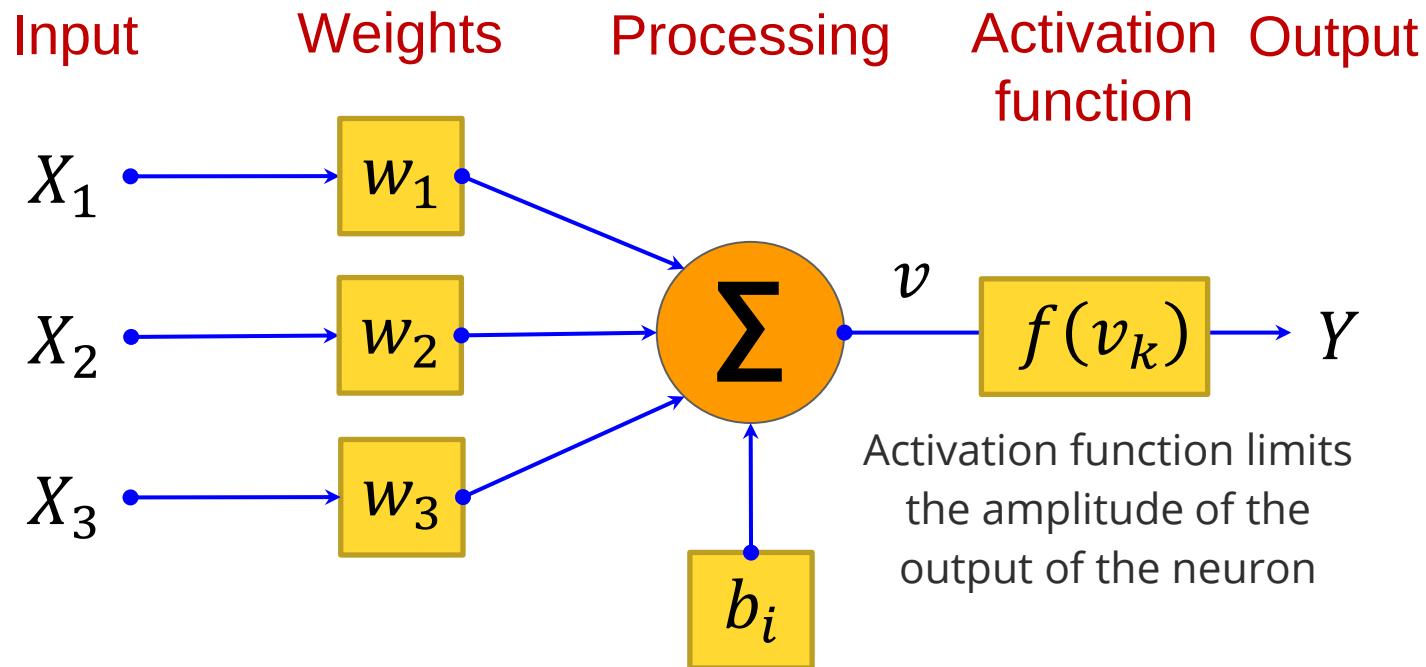
Neural Network

- A hardware implementation or a computer program machine learning approach inspired by the way in which brain performs a particular learning task.
- When 2 connected neurons are firing at the same time, the synapses strength between them increases.
- Synapses change size and strength with experience.



Neural Network

The output is a function of the input, that is affected by the weights, and the transfer functions.



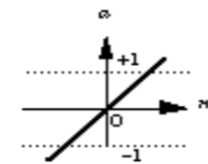
Bias can be incorporated as another weight and makes the neuron more powerful

$$\Sigma = X_1 + X_2 + X_3 = Y$$

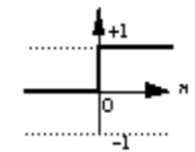
$$\Sigma = X_1 w_1 + X_2 w_2 + X_3 w_2 = Y$$

$$\Sigma X_i w_i = Y$$

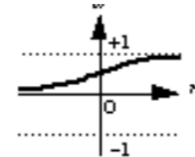
$$Y = f(v_k)$$



Linear



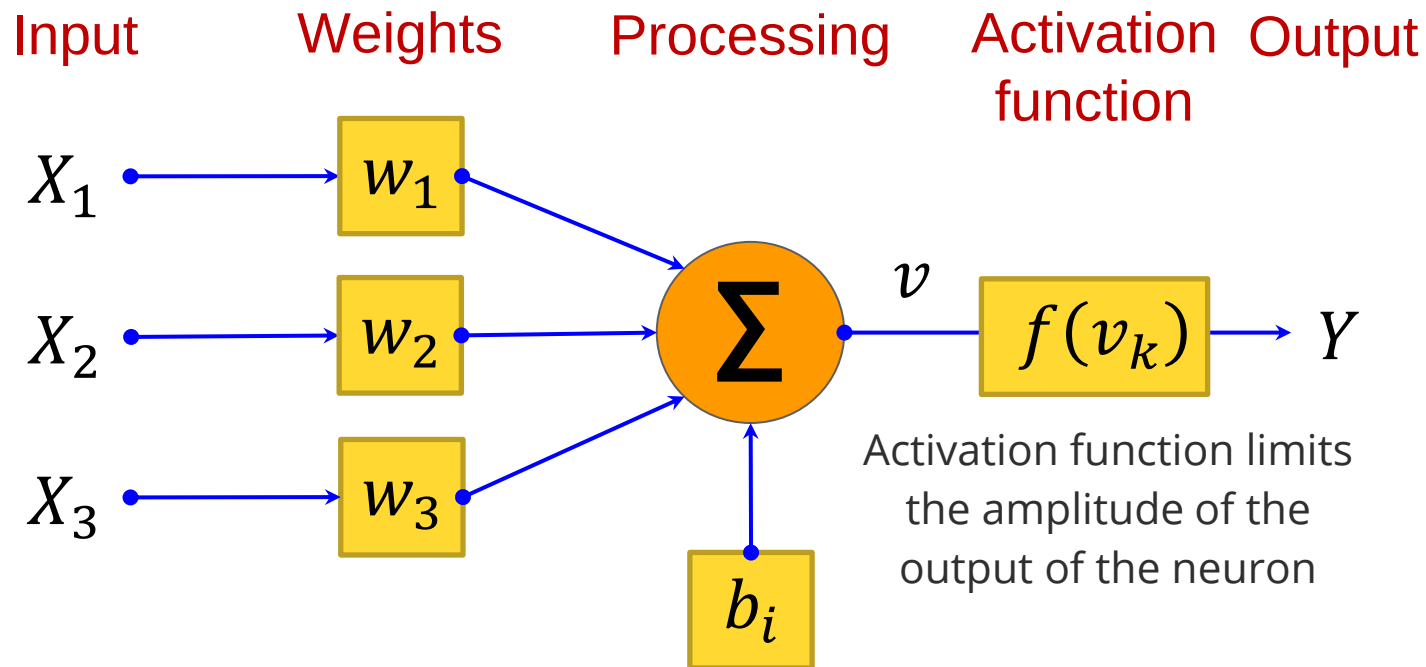
Threshold



Sigmoid

Neural Network

The output is a function of the input, that is affected by the weights, and the transfer functions.



Bias can be incorporated as another weight and makes the neuron more powerful

In matrix form:

$$v = X \cdot W + b$$

$$v = [x_1 \ x_2 \ x_3] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} + b$$

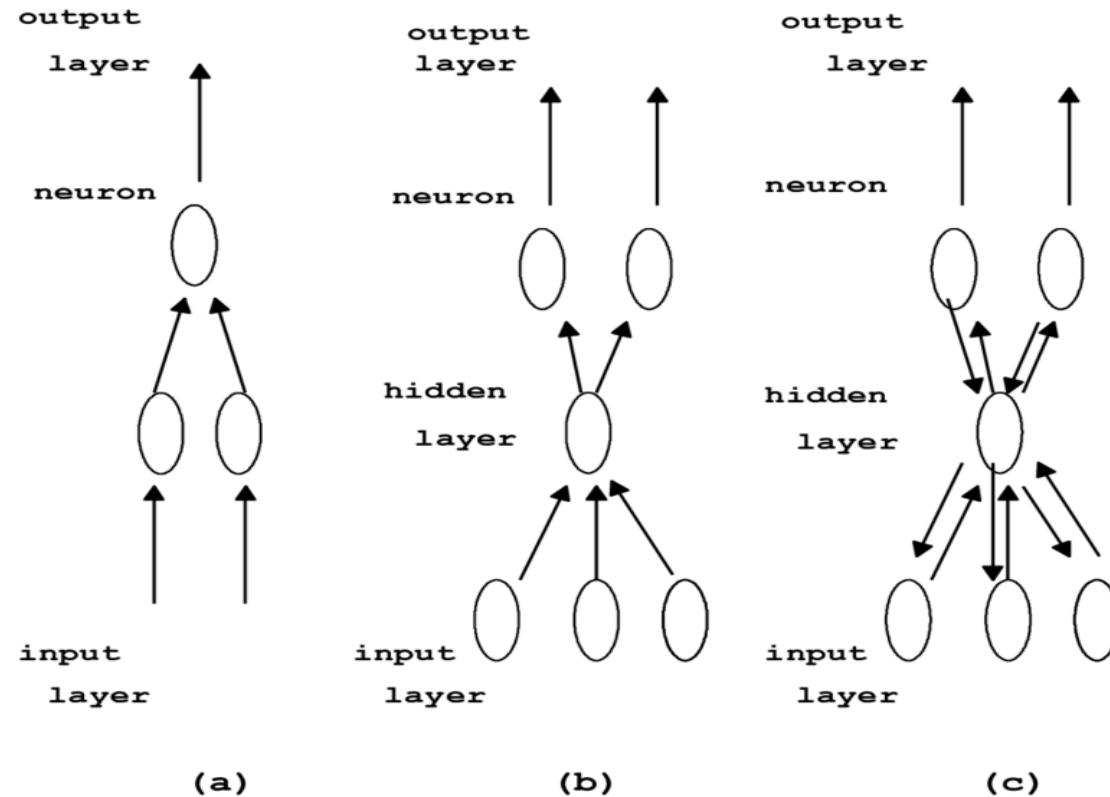
Diagram labels for the matrix equation:

- w_1, w_2, w_3 are grouped and labeled as **weight**.
- x_1, x_2, x_3 are grouped and labeled as **input**.
- b is labeled as **bias**.

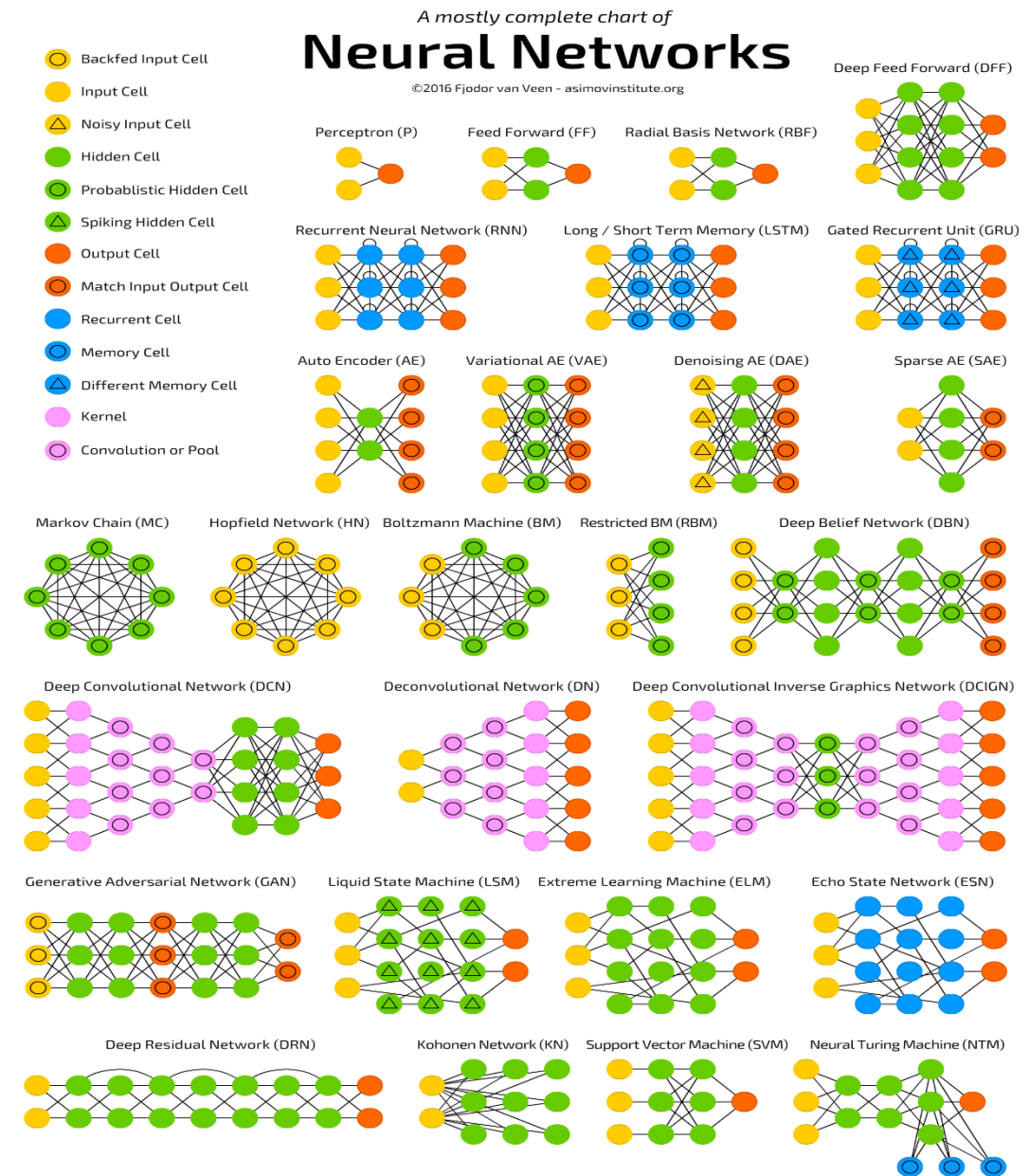
$$Y = \varphi(v)$$

Activation function

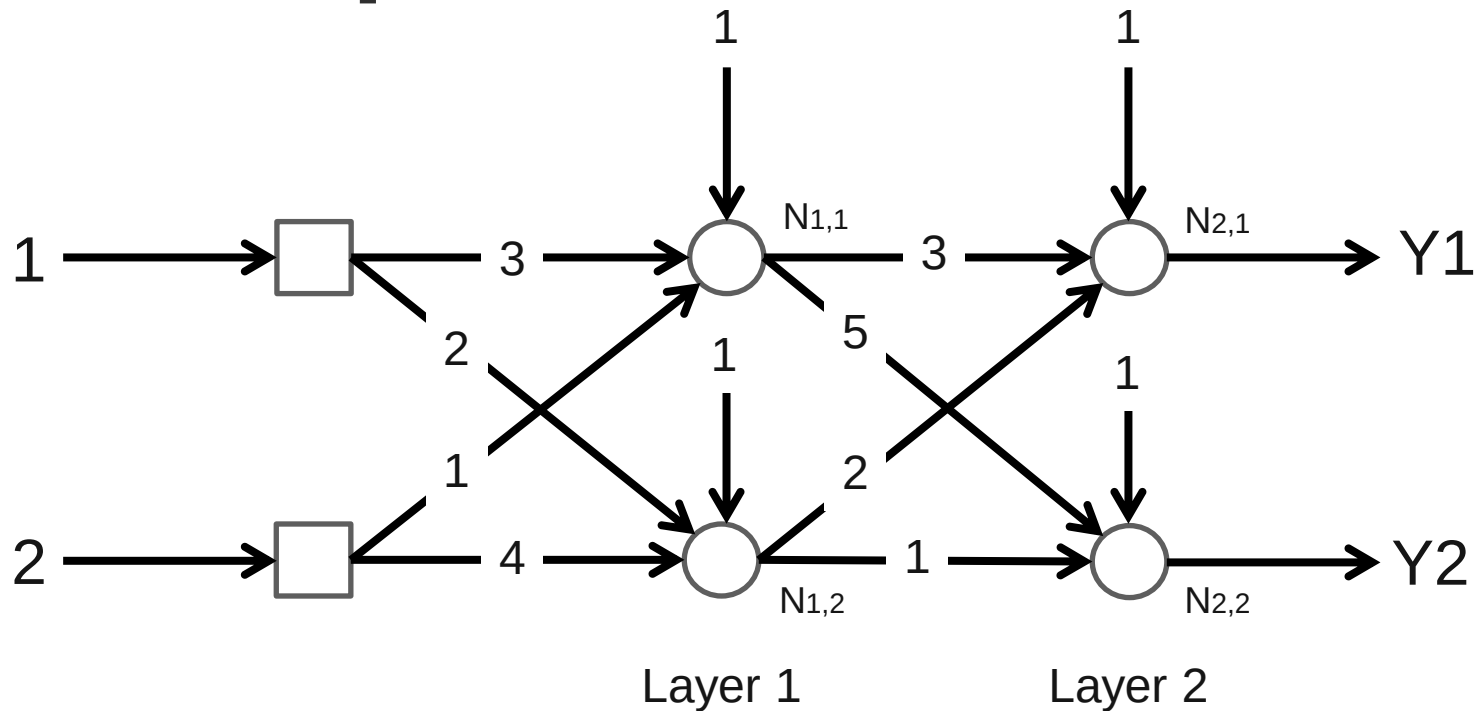
Topologies



(a) single-layer feedforward (b) multilayer feedforward;
(c) multilayer recurrent



Example



$$V_{L1} = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}$$

$$V_{L2} = \begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 41 \\ 42 \end{bmatrix}$$

$$Y = \varphi(V_{L1})$$

$$Y = \varphi \begin{pmatrix} 41 \\ 42 \end{pmatrix}$$

For φ = linear function, then

$$Y = \begin{pmatrix} 41 \\ 42 \end{pmatrix}$$

NN Learning : (1) Supervised

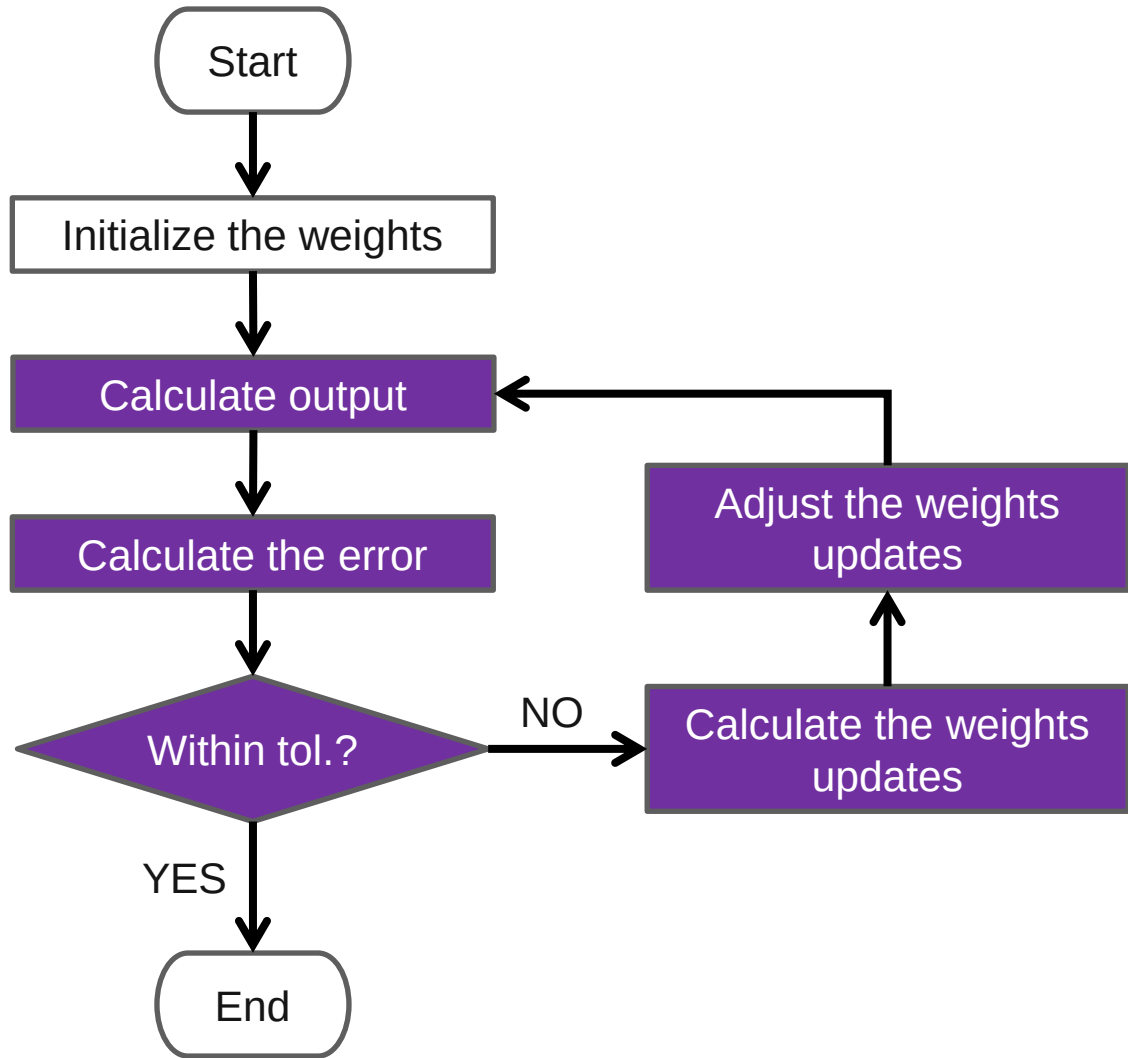
- Fed with inputs and desired output training data.
- NN uses these data to learn the general relationship between the inputs and the outputs.
- This relationship is represented by the values of the weights of the trained network.
- Example: hand-written digits, pattern recognition, regression.
- NN models: perceptron, feed-forward etc.

NN Learning : (2) Unsupervised

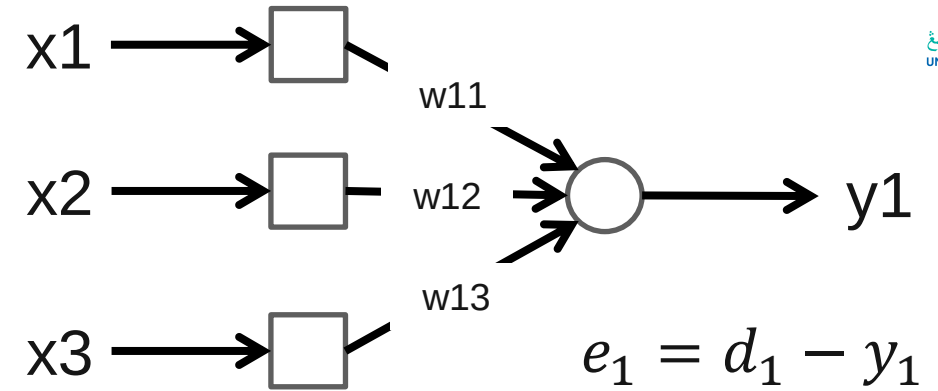
- No desired output is associated with the training data.
- Used to find out structures within data or similar groups.
- Example: find similar groups of documents in the web, clustering & compression.
- NN models: self organizing map, Hopfield networks.

Learning Rule for Supervised Learning

- A systematic approach to modify the weights using backpropagation training method.
- Backpropagation algorithm is to learn for this input output by reducing the error.
- The output error difference between actual and expected results is calculated.
- Backpropagation is fast, flexible, simple and easy to program.



* Epoch = Training iteration



Δ Rule, e

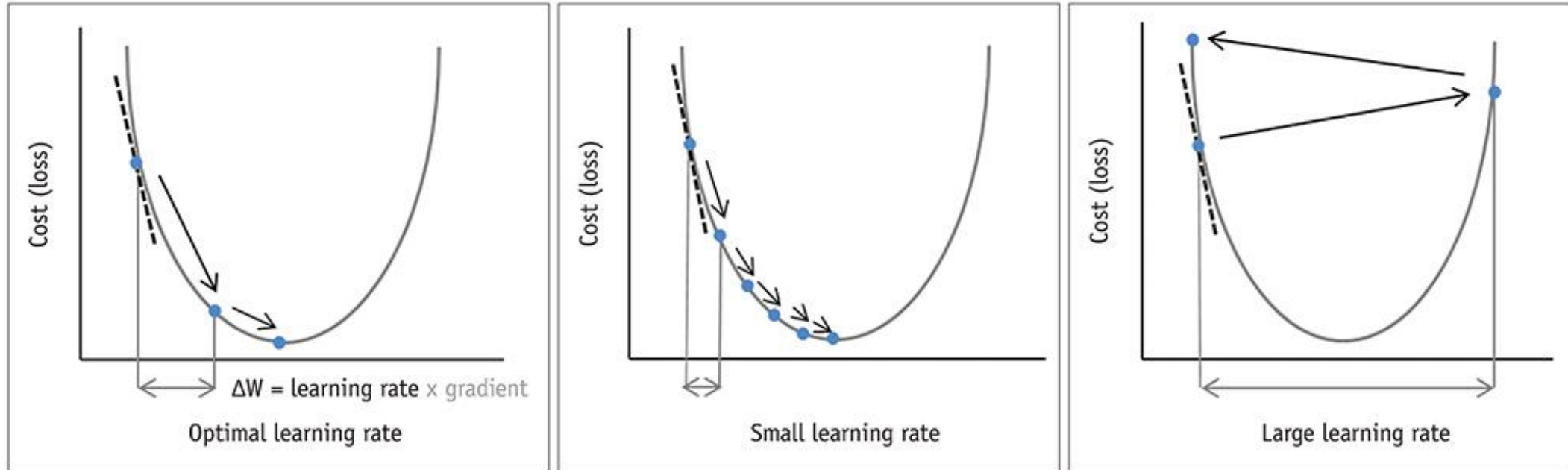
$$w_{1,1} \leftarrow w_{1,1} + \alpha e_1 x_1$$

$$w_{1,2} \leftarrow w_{1,2} + \alpha e_1 x_2$$

$$w_{1,3} \leftarrow w_{1,3} + \alpha e_1 x_3$$

α is learning rate

Learning rate



- Determines the size of the jumps your model makes and, as a result, how rapidly it learns.
- Learning rate too low will take a long time for the ANN model to converge.
- High learning rate on the other may not lead to convergence.

Generalized Δ Rule, δ

$$w_{ij} \leftarrow w_{ij} + \alpha \delta_i x_j$$

$$\delta_i = \varphi'(v_i) e_i$$

$$\varphi' = \frac{d}{dx} (\text{activation function})$$

If linear function, then $\varphi' = 1$

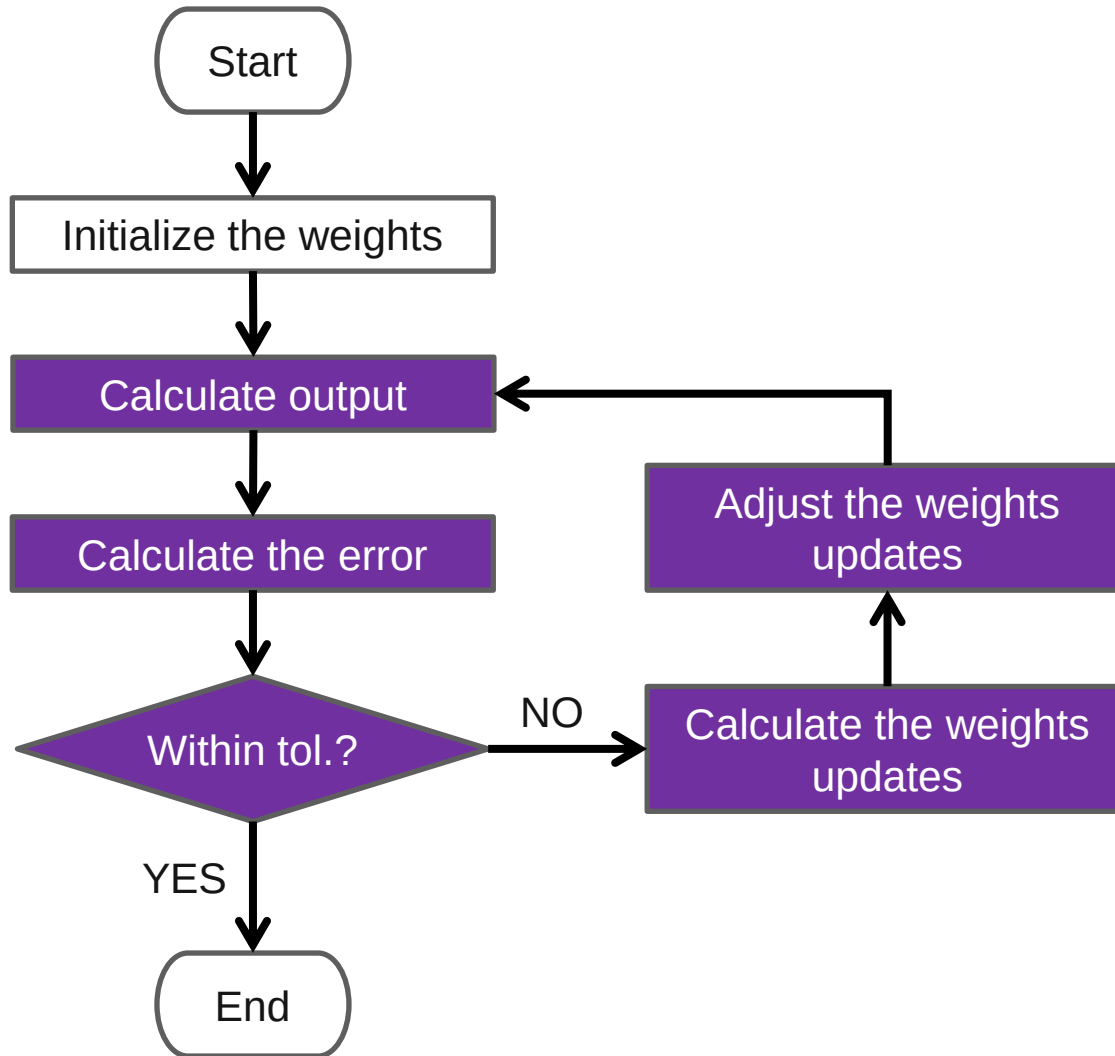
$$\delta_i = (v_i) e_i$$

If sigmoid function,

$$\varphi' = \frac{d}{dx} \left(\frac{1}{1 + e^{-x}} \right)$$

$$\delta_i = [\varphi(v_i)(1 - \varphi(v_i))](v_i) e_i$$

$$w_{ij} \leftarrow w_{ij} + \alpha [\varphi(v_i)(1 - \varphi(v_i))](v_i) e_i x_j$$



* Epoch = Training iteration

Tolerance for Performance

Mean square error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values

- It is always a positive value that decreases as the error approaches zero.

Sum square error (SSE)

$$RSS = \sum_{i=1}^n (y_i - f(x_i))^2$$

RSS = residual sum of squares

y_i = i^{th} value of the variable to be predicted

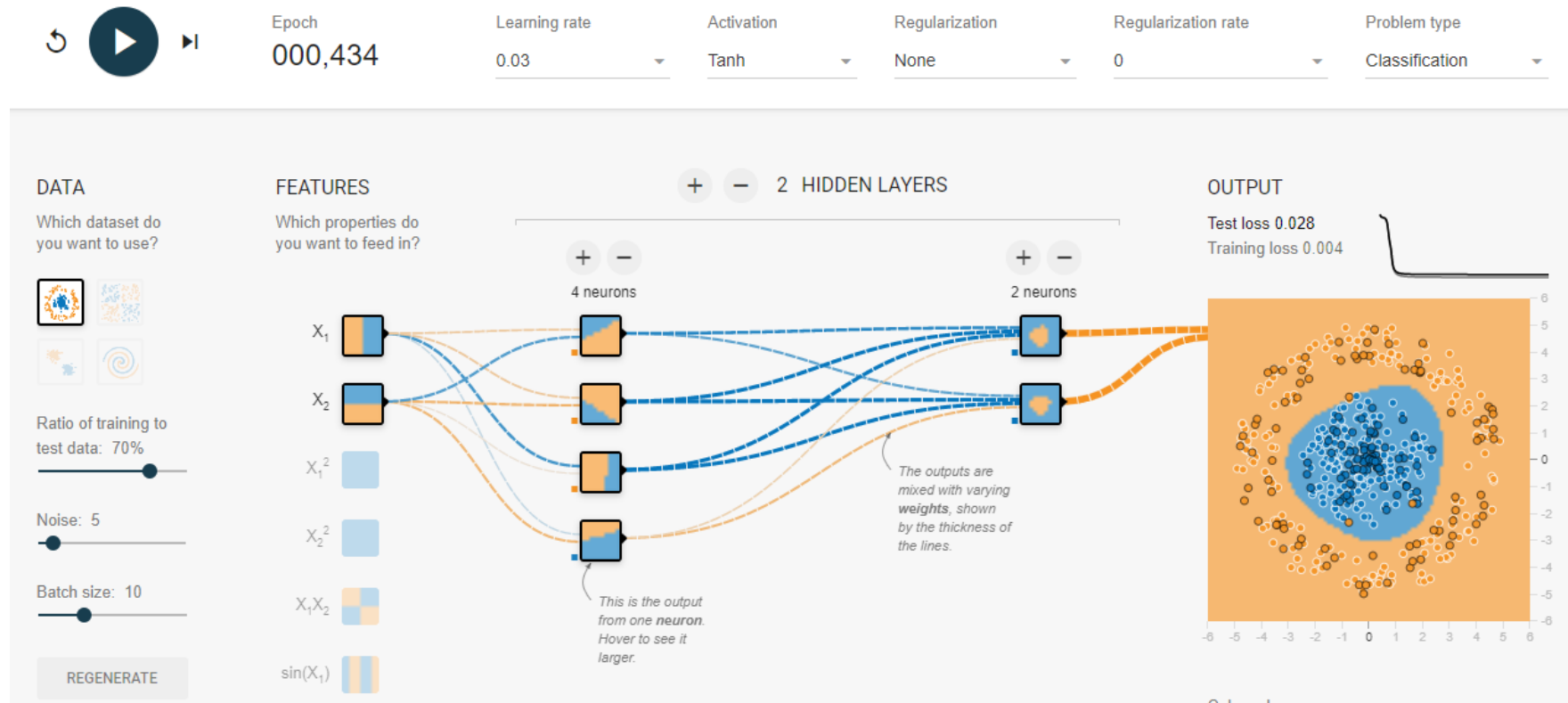
$f(x_i)$ = predicted value of y_i

n = upper limit of summation

- A small RSS indicates a tight fit of the model to the data.

ANN Example in TensorFlow

Tinker With a **Neural Network** Right Here in Your Browser.
Don't Worry, You Can't Break It. We Promise.



Exercise

Analyse the output results for typical classification problem using the following ANN base case (BC) setup.

- Input type: Circle
- Activation method: ReLU
- Regularization: None
- Regularization rate: 0

Analyse the output results when input using different network setting.

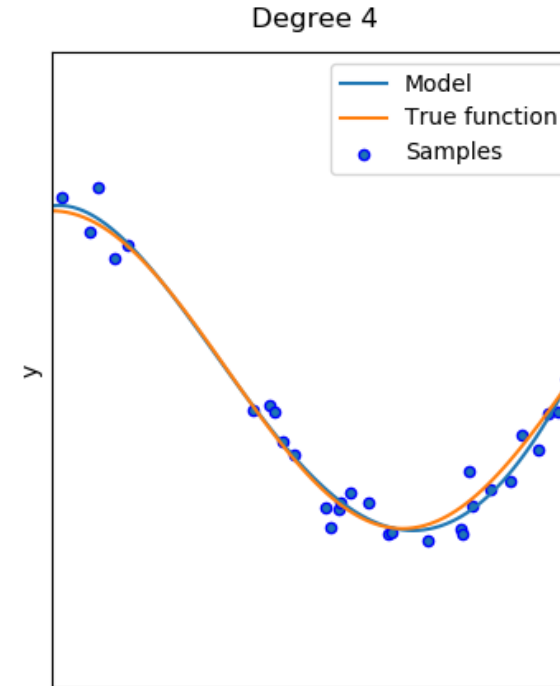
1. BC using activation method Sigmoid
2. BC without regularization

Training Remarks

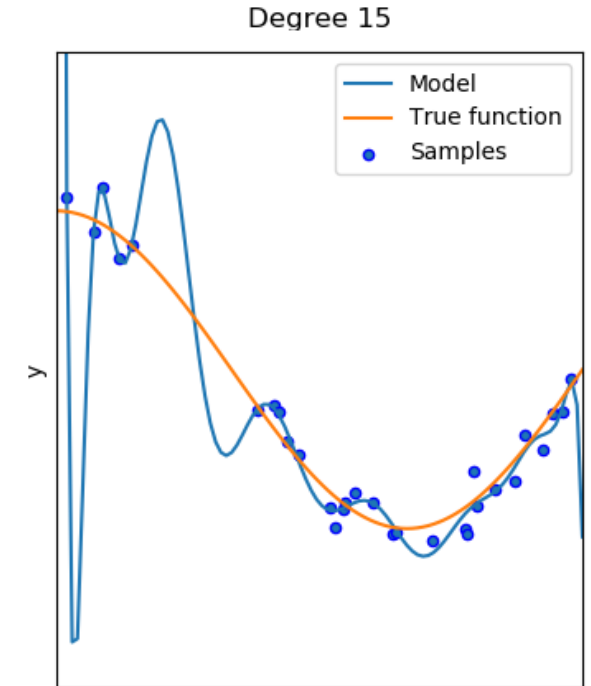
- Backpropagation training is like “Shoe Lace”
- Too little tension - Not enough constraining and very loose
- Too much tension - too much constraint (overtraining), taking too much time (relatively slow process), higher likelihood of breaking
- Pulling one lace more than other - Discomfort (bias)

Training Remarks - Overfitting

- Overfitting is caused by noise in the training data.
- High variance (overfitting) may worsen the ANN performance.
- In practice, overfitting causes the neural network model to perform very well during training, but the performance gets much worse during inference time when faced with brand new data.



Good Fit



High Variance

Training Remarks - Overfitting

- Less complex neural networks are less susceptible to overfitting.
- Running too many epochs or hidden layers can result in over fitting.
- Too few hidden units prevents from adequately fitting the data.
- To prevent overfitting or a high variance we must use something that is called regularization.
- Regularization techniques: L1 & L2
 - Regularization rate is too high, run the risk of underfitting the data.
 - Regularization rate is too low, run the risk of overfitting the data.

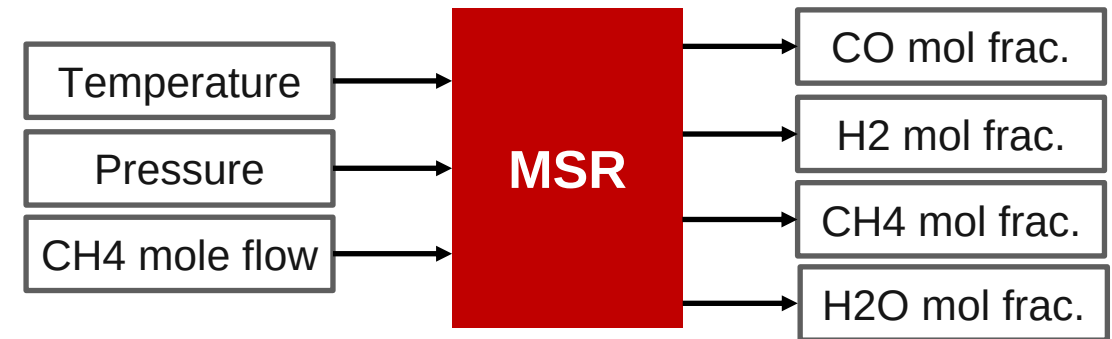
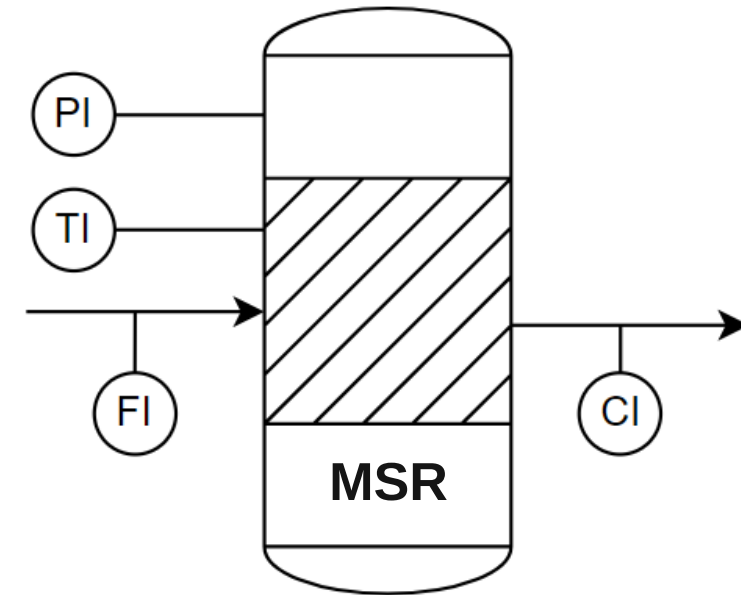
MATLAB ANN Model

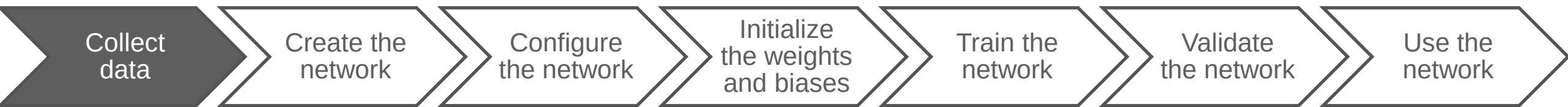
1. Collect data
2. Create the network
3. Configure the network
4. Initialize the weights and biases
5. Train the network
6. Validate the network
7. Use the network

Example: ANN Model for H₂ Production via MSR

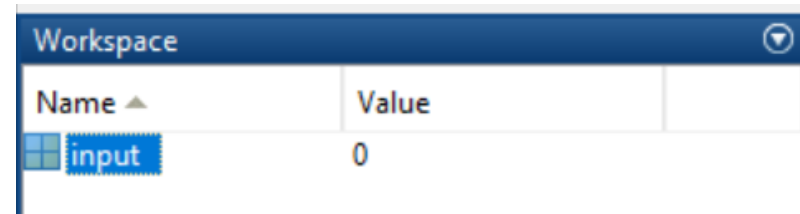
Methane Steam Reforming (MSR) is an important process for hydrogen production and the hydrogen economy. This process converts methane (from natural gas) and water into hydrogen and carbon dioxide. Many simulations of the MSR process are either done as balance-based, equilibrium-based or kinetic-based simulations. These models require a 1st principles MESH equations. However, plant data can be used to model the process. This example demonstrates how build and deploy an ANN model using historical plant data in MATLAB.

Refer to file `SMR_Data_ANN.xls`. The independent input process variables are reactor temperature (TI), reactor pressure (PI) and CH₄ mole flow (FI) whereas the dependant output variables (CI) are the output stream mole frac for CO, H₂, CH₄ and H₂O.



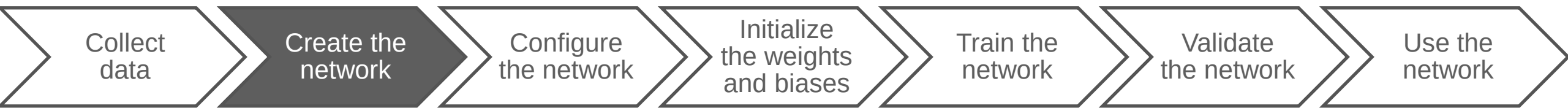


1. In Workspace environment right click and select New to declare new variable. Name it as 'input'.
2. Double click the 'input' variable to open the variable editor.
3. In the excel file highlight all of the 3-input data.
4. Right click on the MATLAB data editor and click Paste Excel Data to insert the data.
5. The input/data format is based on column. Therefore, the original data need to be transposed. Right click on the editor and click Transpose Variable.
6. 'input' variable should be in 3x100 double format.
7. Repeat the steps for the 4-output data.

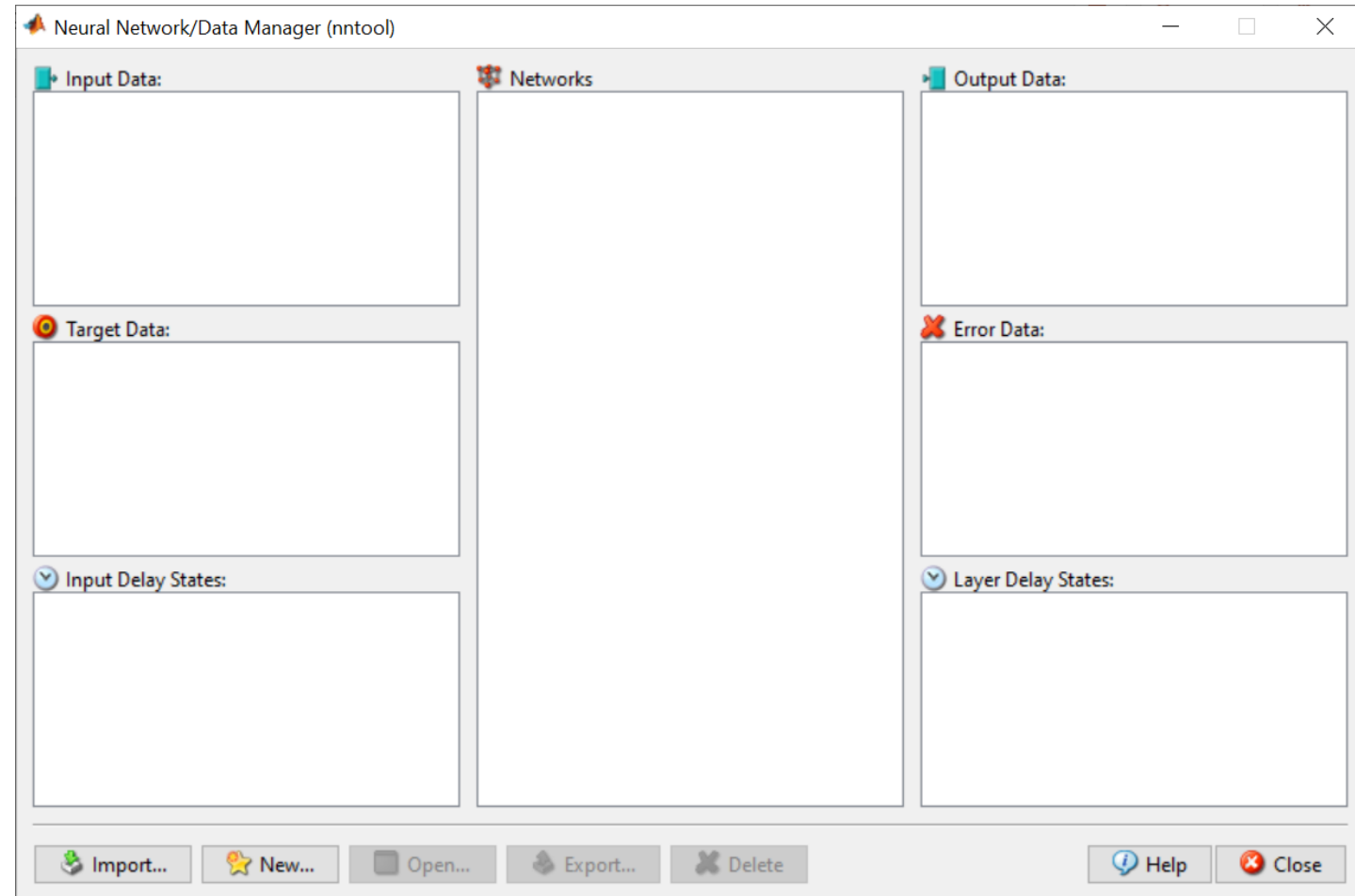


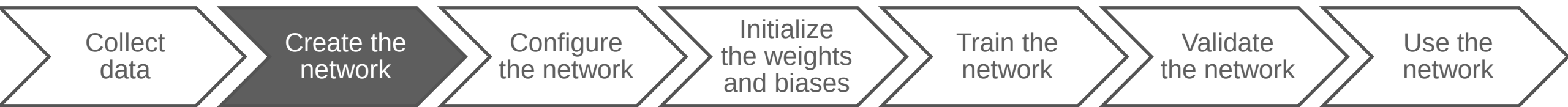
	1	2	3	4	5	6	7	8	9	10	11	12
1	1.5965e+03	1632	1372	1.4603e+03	1.8105e+03	1.6575e+03	1.3383e+03	1.3494e+03	1.6796e+03	1813	1.4827e+03	1.4811e+03
2	416.6250	326.6100	505.3650	393.2240	520.7560	303.2120	429.0240	350.0730	523.3200	292.2000	421.1060	310.2700
3	232.3200	347.8200	144.4300	285.6700	88.8250	186.7250	374.8800	176.1144	351.1411	264.0607	65.3334	419.3803
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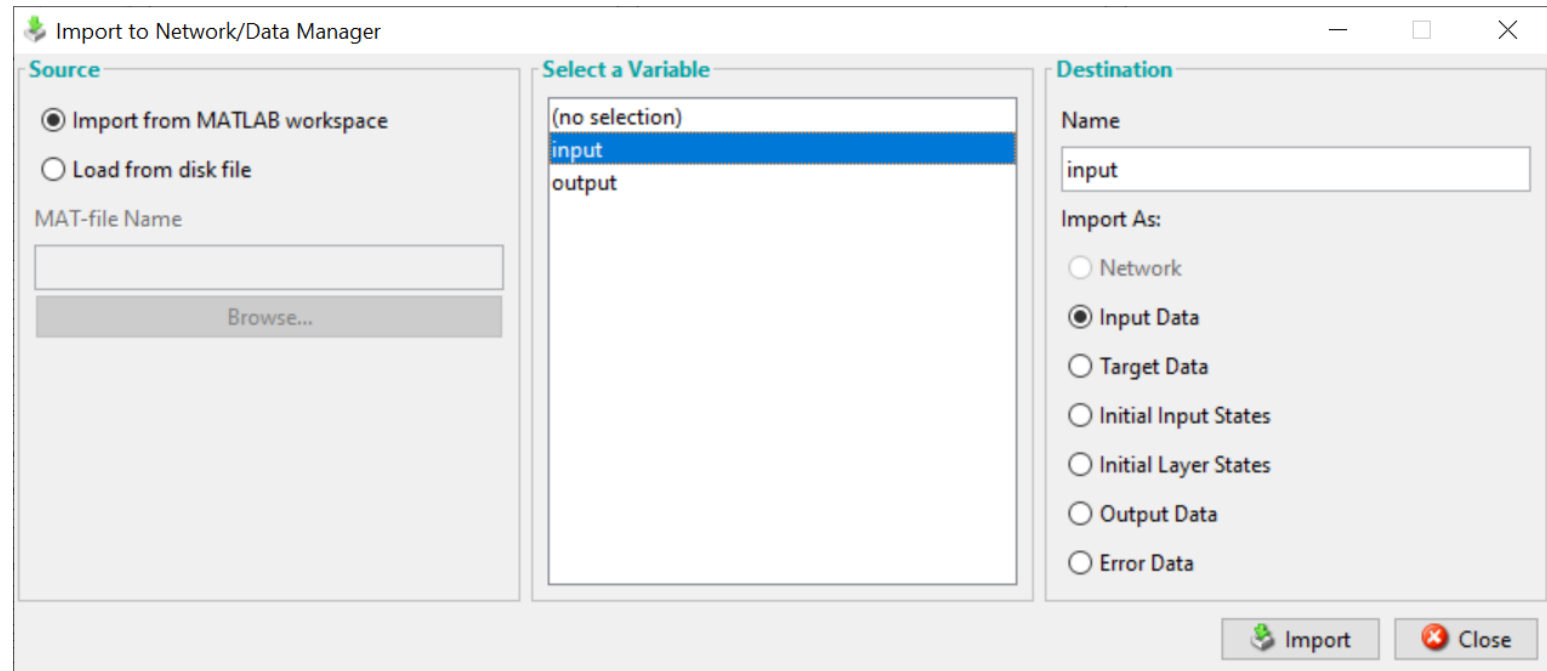


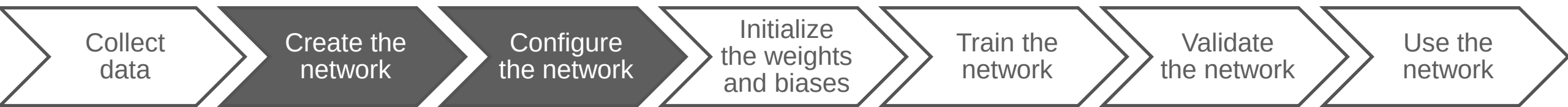
1. In MATLAB Command Window type 'nntool' to open the Neural Network Tool GUI.
2. To import the input/output data click Import to open the Import to Network/Data Manager.





1. In the Network/Data Manager window, choose Import from MATLAB workspace in the Source section
2. In the Select a Variable section select the 'input' variable and select Input Data as the Destination.
3. Click Import and click OK.
4. Repeat the same step for 'output' variable and select Target Data as Destination.
5. The 'input' and 'output' variable will be declared in the Input Data and Target Data section of the Network/Data Manager.





1. To create a network, in the nntool GUI click New to open Create Network or Data window.
2. The default network name is given. You can rename it accordingly.
3. In the Network Properties section select the 'input' and 'output' variable as Input data and Target data respectively.
4. Here you can choose other network properties but for this exercise remain as it is.
5. Click Create to declare the network in the Network/Data Manager.

Create Network or Data

Network Data

Name: network1

Network Properties

Network Type: Feed-forward backprop

Input data: (Select an Input)

Target data: (Select a Target)

Training function: TRAINGD

Adaption learning function: LEARNGDM

Performance function: MSE

Number of layers: 2

Properties for: Layer 1

Number of neurons: 10

Transfer Function: TANSIG

View Restore Defaults

Help Create Close

Collect
data

Create the
network

Configure
the network

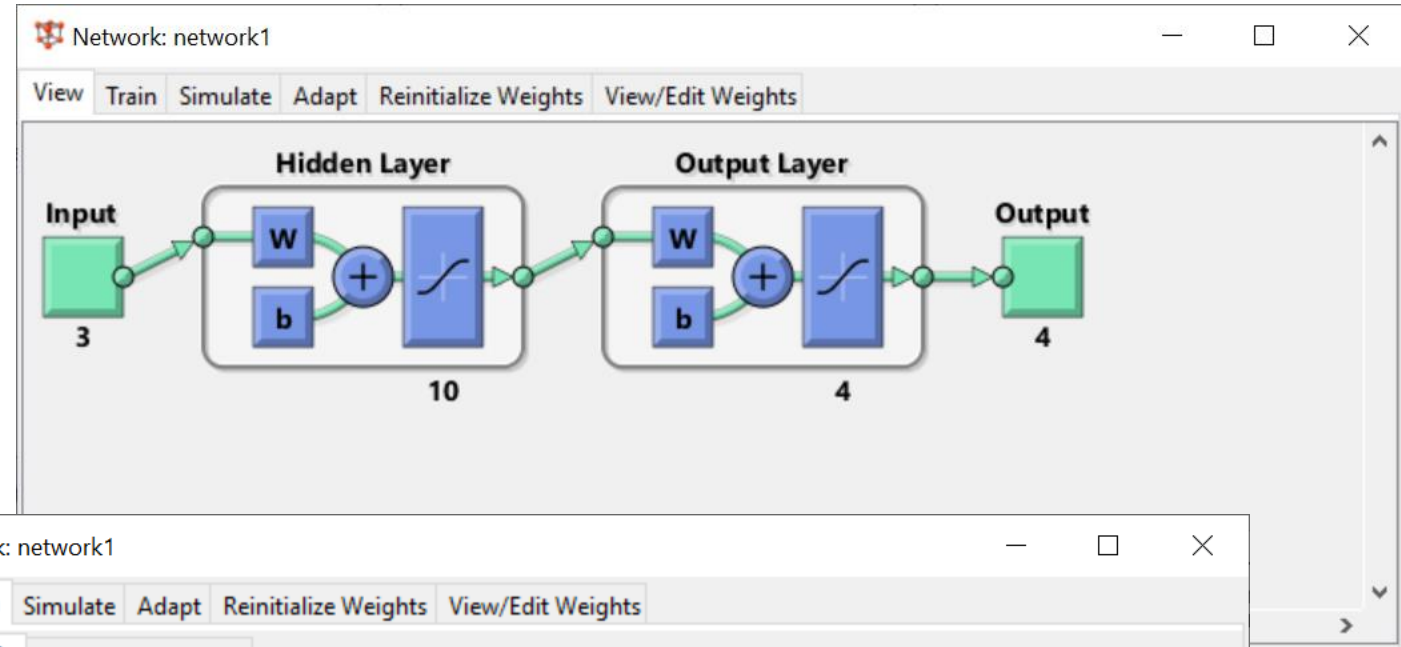
Initialize
the weights
and biases

Train the
network

Validate
the network

Use the
network

1. To configure the network, double click 'network1' in the nntool GUI to open the network configuration window.
2. Go to Train tab and select the 'input' and 'output' variable as Inputs and Targets data respectively.
3. To start the training, click Train Network in the network configuration window.



Network: network1

View Train Simulate Adapt Reinitialize Weights View/Edit Weights

Training Info Training Parameters

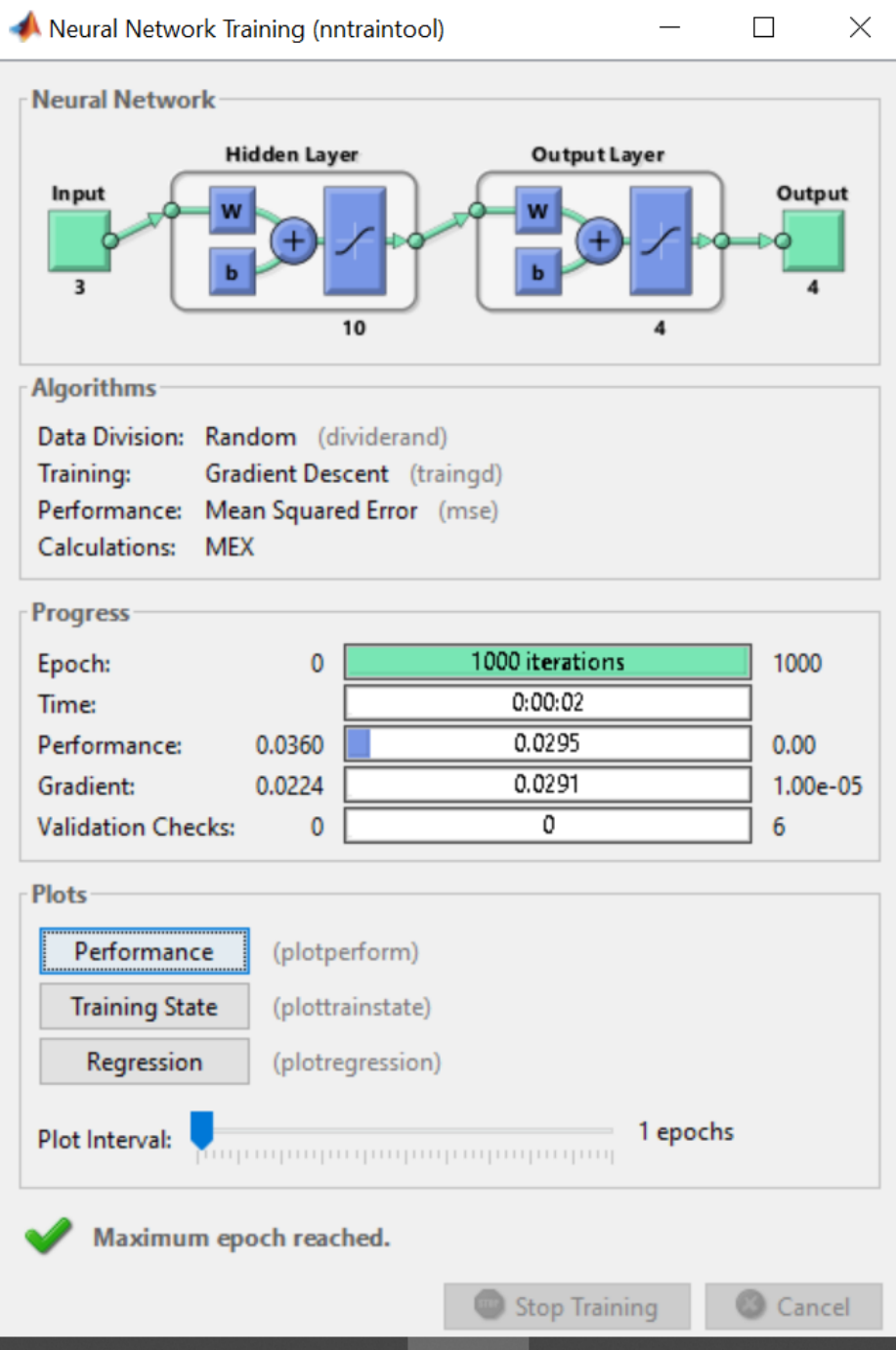
Training Data

Inputs	input
Targets	output
Init Input Delay States	(zeros)
Init Layer Delay States	(zeros)

Training Results

Outputs	network1_outputs
Errors	network1_errors
Final Input Delay States	network1_inputStates
Final Layer Delay States	network1_layerStates

Train Network



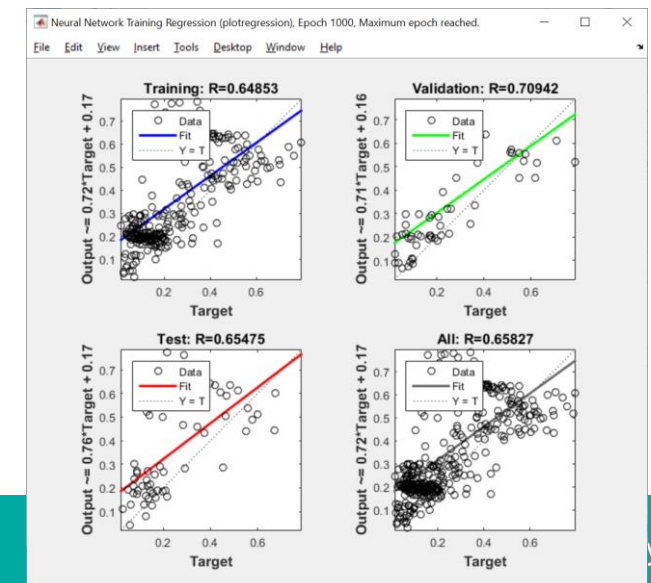
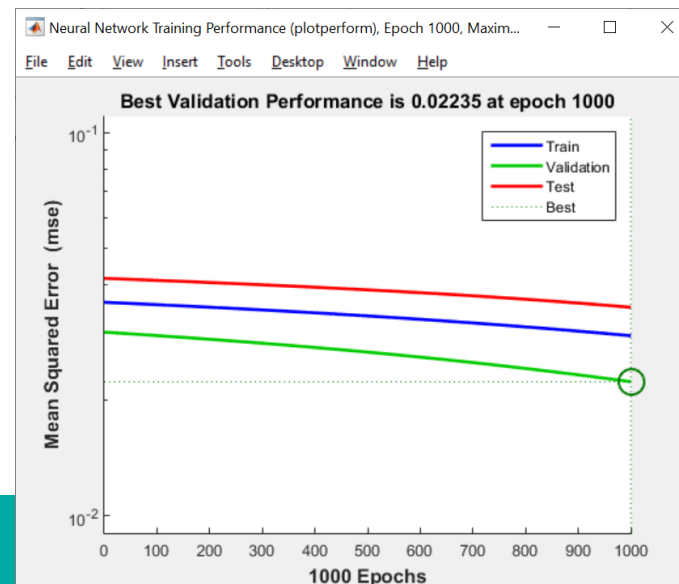
Initialize
the weights
and biases

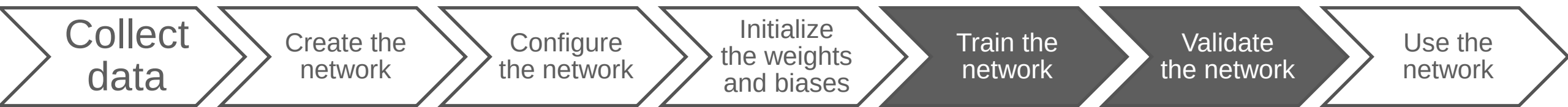
Train the
network

Validate
the network

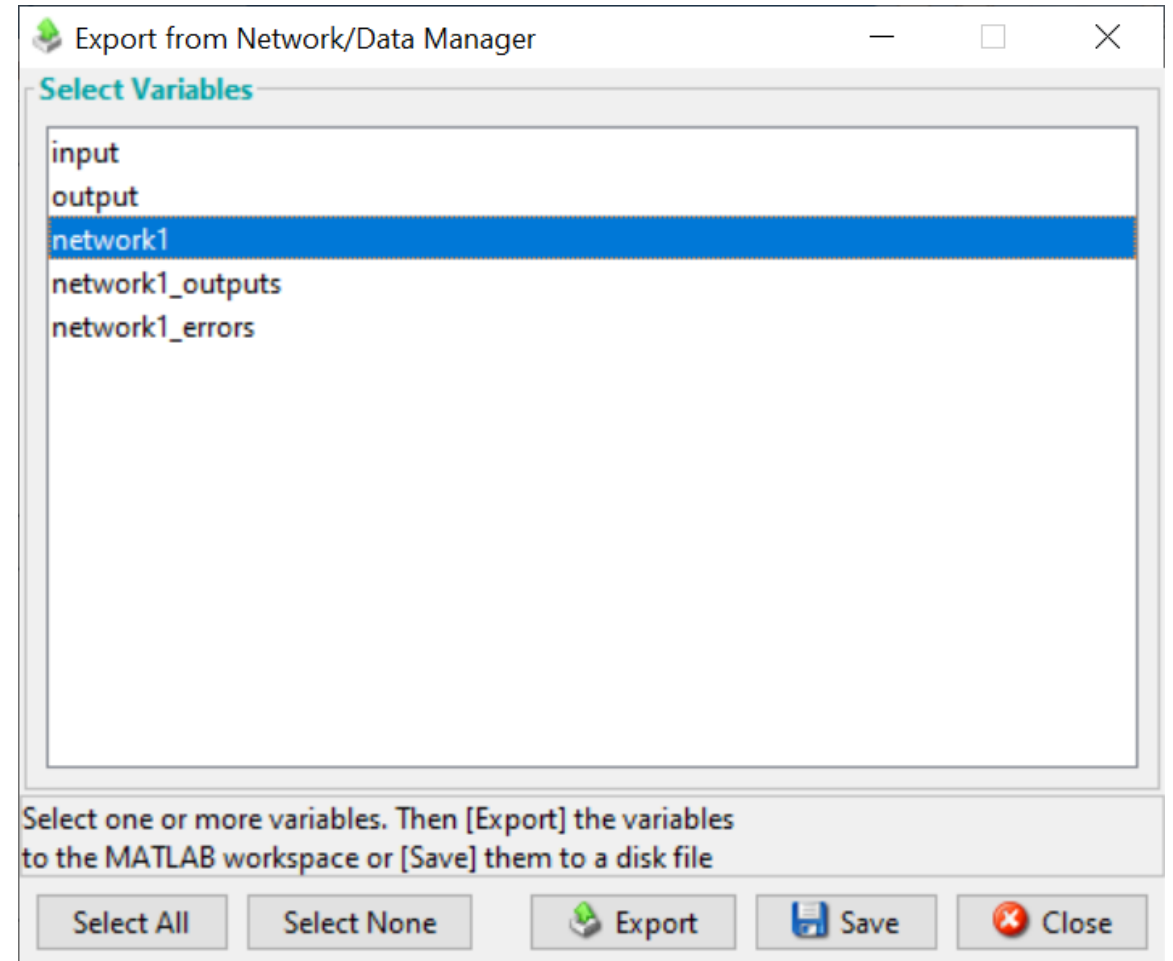
Use the
network

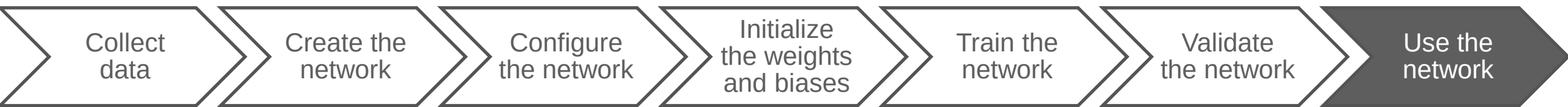
1. The Neural Network Training (nntraintool) window shows the NN topology, training status and performance plots.
2. In the Plots section, you can view the Performance, Training State and Regression graph.
3. The Regression plot shows the performance plot for Training, Validation, Test and All.





1. To use the network, it need to be exported to the Workspace.
2. In the nntool GUI window click Export to open the Export from Network/Data Manager window.
3. In the Select Variables section, select network1 and click Export to export the network to MATLAB Workspace
4. Once declared the model is ready to be tested.





	A	B	C	D	E	F	G
1	001:A1005.PV	001:A1006.PV	001:A1007.PV	001:A1008.PV	001:A1009.PV	001:A1010.PV	001:A1011.PV
2	Reactor Temperature	Reactor Pressure	CH4 mole flow	CO Mole Frac	H2 Mole Frac	CH4 Mole Frac	H2O Mole Frac
3	F	psia	lbmol/hr	fract.	fract.	fract.	fract.
4				CO	H2	CH4	H2O
5	Value	Value	Value	Value	Value	Value	Value
6	1596.5	416.625	232.32	0.145851649	0.450877373	0.098187872	0.307480113

1. To test the network prediction capability, first create a test data. The data to be tested is the first set of data in the actual dataset.
2. In the Command Window type the following command
`>> test1 = [1596.5;416.625;232.32]`
3. This will create a new 3x1 variable named 'test1'.
4. Next to test the network, in the Command Window type the following command
`>> calc_output= sim(network1,test1)`
5. Compare it with the actual output data.

Exercise

Based on the NN model example, try different network sizes and different teaching algorithms. Repeat the exercise with the following network configuration and compare the results with the base case (BC) or initial NN model. Note that for each run you need to create a new network.

Run	Training function	No. of neurons	Transfer function
BC	TRAINGD	10	TANSIG
1	TRAINGD	15	TANSIG
2	TRAINGD	10	LOGSIG
3	TRAINLM	10	TANSIG
4	TRAINLM	15	LOGSIG

Table: Regression Performance

Run	Training	Validation	Test	All
BC				
1				
2				
3				
4				

Table: Test Results

Test data : test1 = [1596.5;416.625;232.32]

Mol fraction	Actual Output	BC		Run 1		Run 2		Run 3		Run 4	
		Output	% Diff	Output	% Diff	Output	% Diff	Output	% Diff	Output	% Diff
CO	0.1459										
H2	0.4509										
CH4	0.0982										
H2O	0.3075										

Limitations

- Hardware dependence: ANN require processors with parallel processing power, in accordance with their structure. For this reason, the realization of the equipment is dependent.
- Unexplained behavior of the network: This is the most important problem of ANN. When ANN produces a probing solution, it does not give a clue as to why and how. This reduces trust in the network.
- Determination of proper network structure: There is no specific rule for determining the structure of artificial neural networks. Appropriate network structure is achieved through experience and trial and error.

Limitations

- Difficulty of showing the problem to the network: ANNs can work with numerical information. Problems must be translated into numerical values before being introduced to ANN. The display mechanism to be determined here will directly influence the performance of the network . This depends on the user's ability.
- The duration of the network is unknown: The network is reduced to a certain value of the error on the sample means that the training has been completed. This value does not give us optimum results

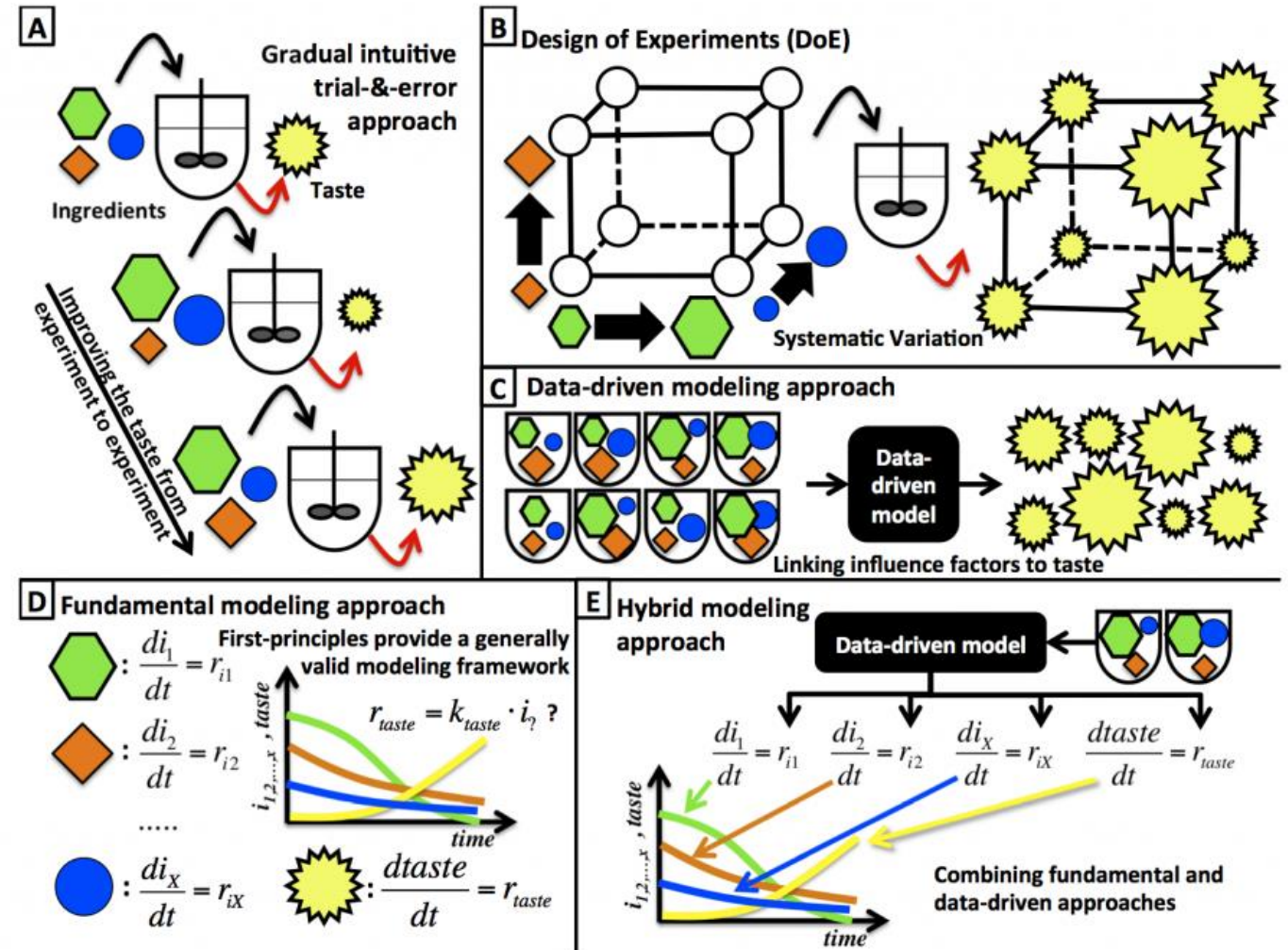
Introduction of HM

- Imagine process of making a cup of coffee. You have all the ingredients, equipment and attempt your first cup.
- How to improve the taste?



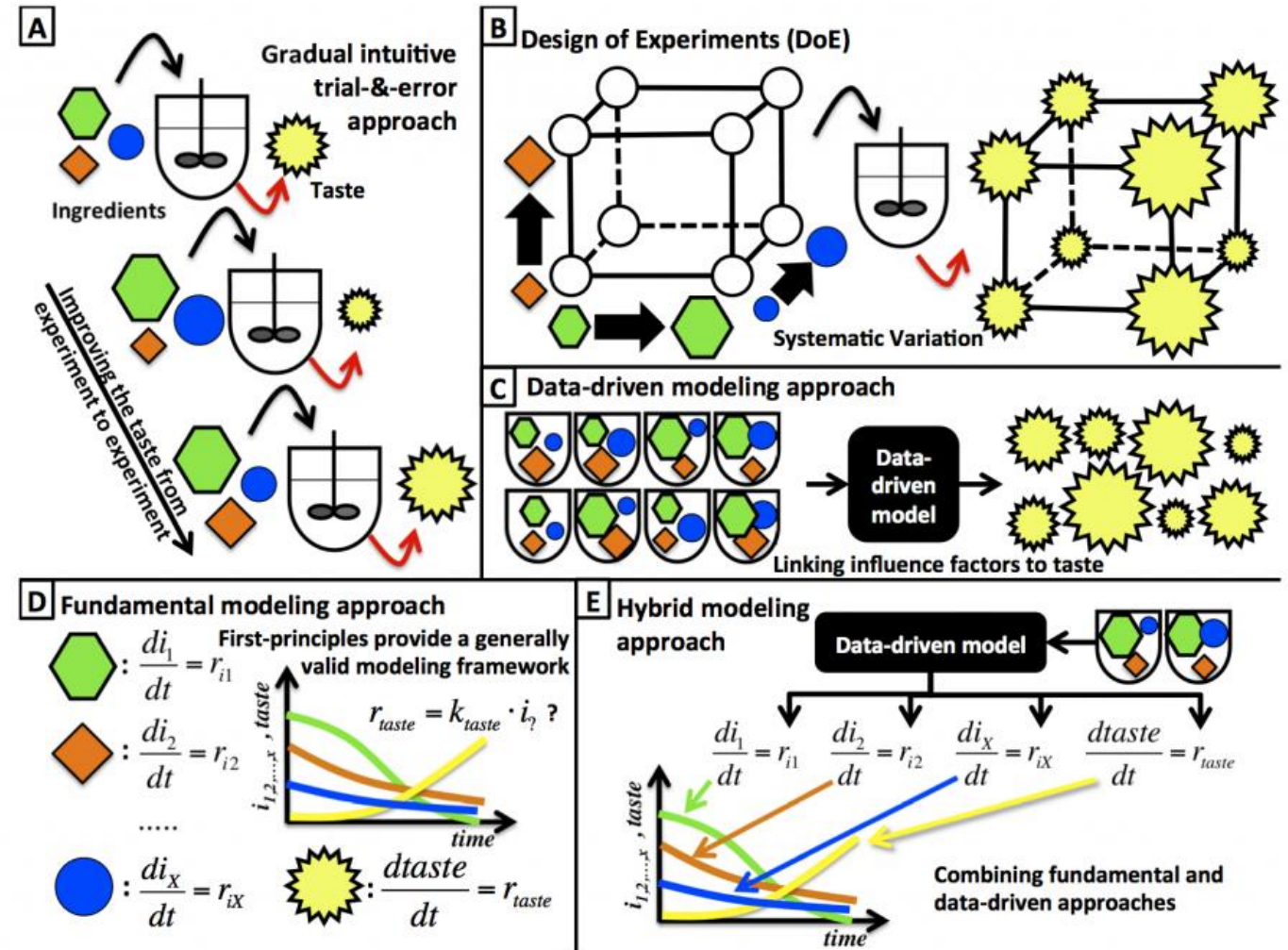
Introduction of HM

- A. Trial and error by adjusting the coffee type, ingredient amount, brewing time, water temperature, pressure etc.



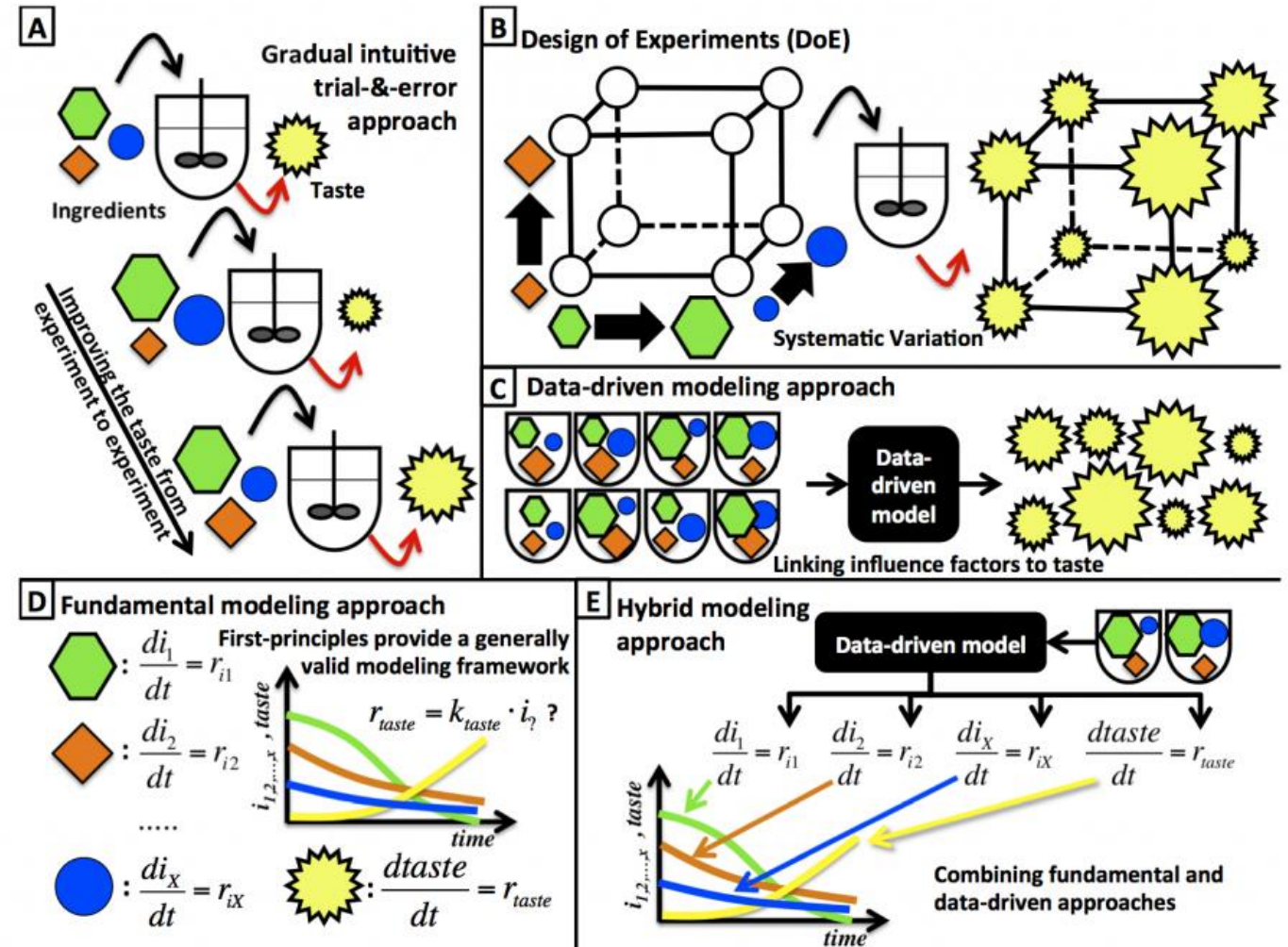
Introduction of HM

- B. Apply statistical design of experiments (DoE). Need to define factors or manipulated variables i.e., mass, water temperature etc. that influence the controlled variables i.e., taste.



Introduction of HM

- C. Quantify the process using measurement equipment i.e., sensors, analyzer etc. With the data, a data-driven model can be developed which links the different influence factors to the analytic quantified taste of coffee.



Introduction of HM

- Approach A, B and C can be performed by anyone and hardly used any fundamental chemical engineering knowledge.
- Putting chemical engineering knowledge to work include physical properties, MEB, reaction rates, diffusion rates etc.

Open Access

Article

Mineral Composition and Antioxidant Potential of Coffee Beverages Depending on the Brewing Method

by  Katarzyna Janda ^{1,*}  ,  Karolina Jakubczyk ¹ ,  Irena Baranowska-Bosiacka ²  ,
 Patrycja Kapczuk ²  ,  Joanna Kochman ¹ ,  Ewa Rębacz-Marón ³  and
 Izabela Gutowska ⁴  

Addition of milk to coffee beverages; the effect on functional, nutritional, and sensorial properties

A. Rashidinejad , O. Tarhan, A. Rezaei, E. Capanoglu, S. Boostani, S. Khoshnoudi-Nia, ...show all

Pages 6132-6152 | Published online: 11 Mar 2021

 Download citation

 <https://doi.org/10.1080/10408398.2021.1897516>

 Check for updates

Effect of brewing technique and particle size of the ground coffee on sensory profiling of brewed Dampit robusta coffee

K Fibrianto¹, Y R Febryana¹ and E S Wulandari¹

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What is HM?

Fundamental Model

- Allow explaining the behavior of a system in a broad range.
- Involve MESH equation systems or mechanistic model
- May not capture all phenomenon
- Tedious
- Time consuming
- Expertise required

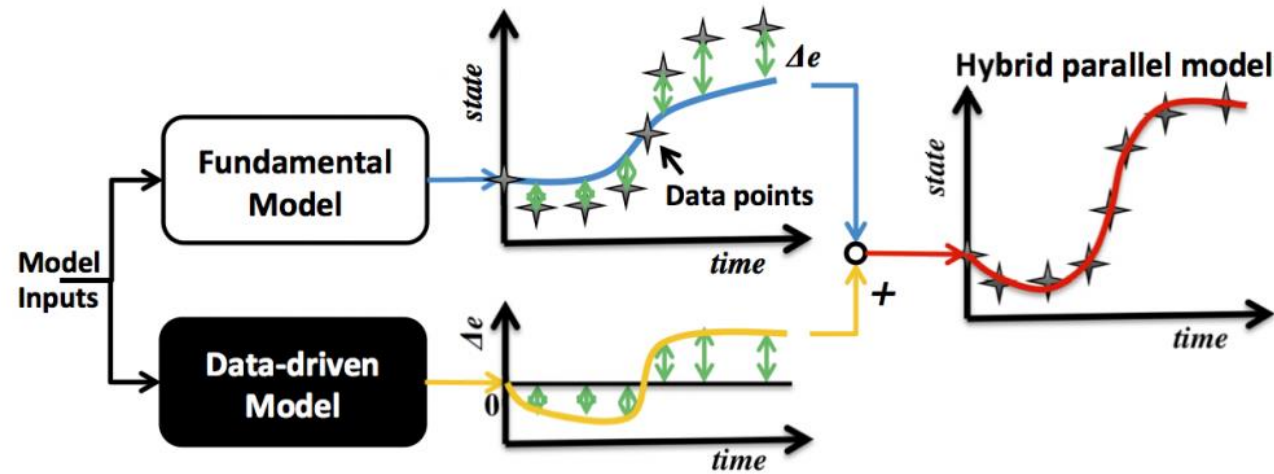
Hybrid Model

- Integrate and capture all available knowledge of 1st principal and process data.
- Less derivation
- Less experiments

Data-Driven Model

- Can be applied when nothing at all is known about the functioning of the system.
- Require a lot of 'good data'
- Difficult to interpret
- Poor extrapolation capability
- May violate physical constraints

Applications



- When different sources of knowledge are available i.e., process model and plant data.
- Suitable for objective drive i.e., a model for optimization.
- Suitable to represent the variables in the fundamental model for which it is difficult to derive mathematical expressions i.e., tray efficiencies, reaction rates, mass transfer rates, friction factors, catalyst deactivation, heat transfer coefficients, performance curve.

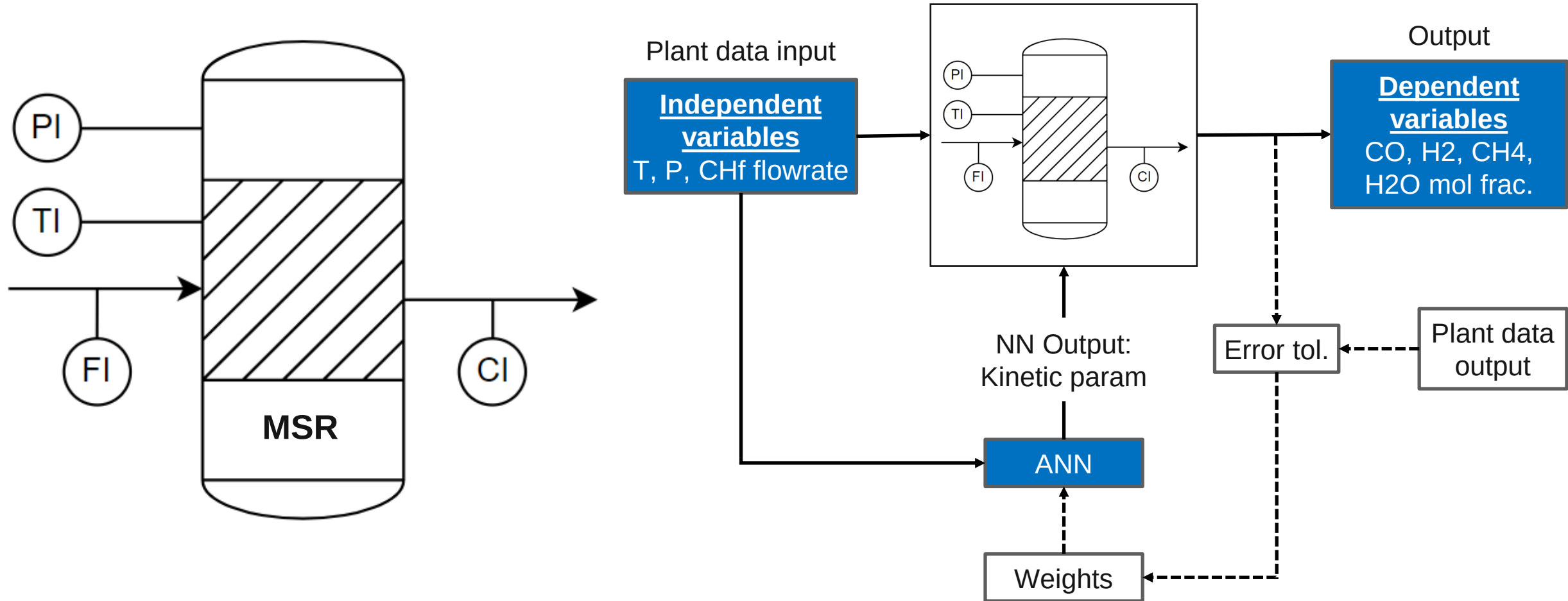
General Steps

1. Build process flowsheet with first principles model
2. Evaluate the accuracy of first principles model
3. Train a hybrid model
4. Deploy the hybrid model

Example: HM for H₂ Production via MSR

Many simulations of the MSR process are either done as balance-based or equilibrium-based simulations. However, kinetics-based simulations are necessary to improve and optimize the reactor performance. In this example, the MSR reactor is modeled in Aspen Plus using the RPLUG block with Langmuir-Hinshelwood-Hougen-Watson (LHHW) kinetics. Plant data can be used to tune and improve the kinetic parameters in the model to better represent real performance. This example demonstrates how to leverage AI training with plant data to evaluate and improve the accuracy of the MSR kinetic reaction model.

Hybrid model architecture

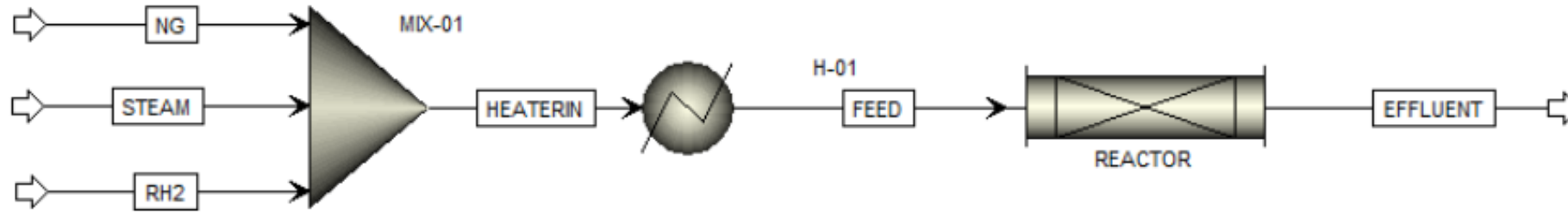


Build process model
flowsheet

Evaluate model

Train HM

Deploy HM



Water, methane, carbon monoxide and hydrogen are defined as components. The thermodynamic method used is RKSMHV2 based on the Redlich-Kwong-Soave equation of state with modified Huron-Vidal mixing rules. The feed stream to the MSR reactor contains natural gas and steam. Both streams are mixed in MIX-01 at 29 atm before being heated up to 700 °C. The diameter of the MSR reactor is defined as 0.6 m and the length as 10 m. After the heater, the process stream goes to the MSR reactor represented by an RPLUG model block operated isothermally. The adsorption constant for H₂O will be regressed with plant data.

Build process model
flowsheet

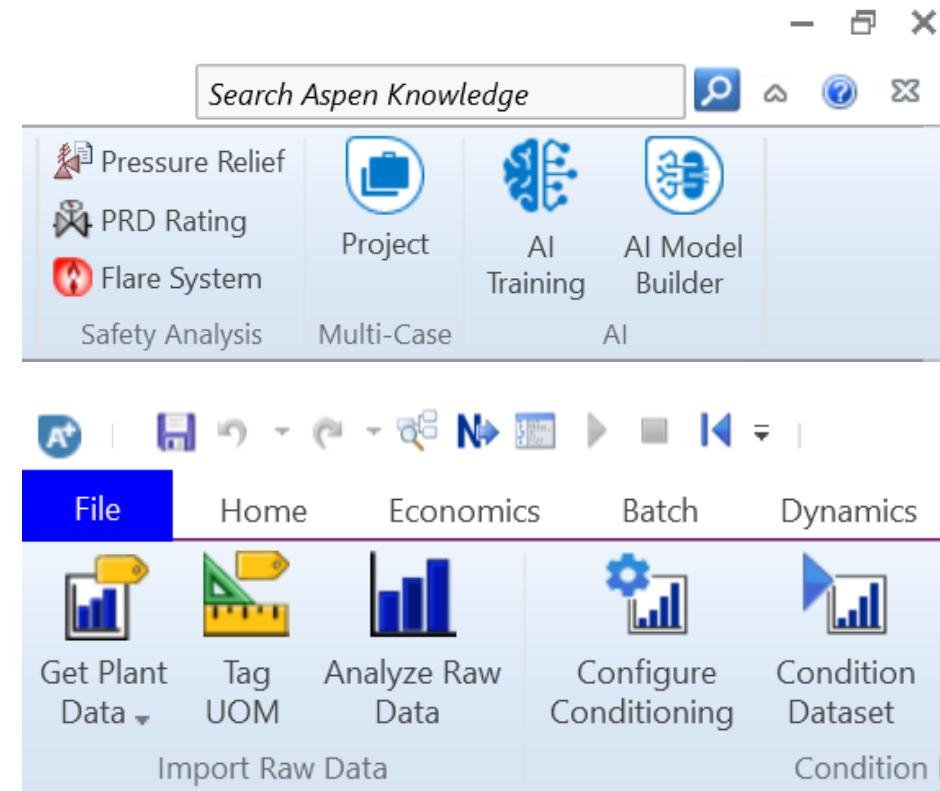
Evaluate model

Train HM

Deploy HM

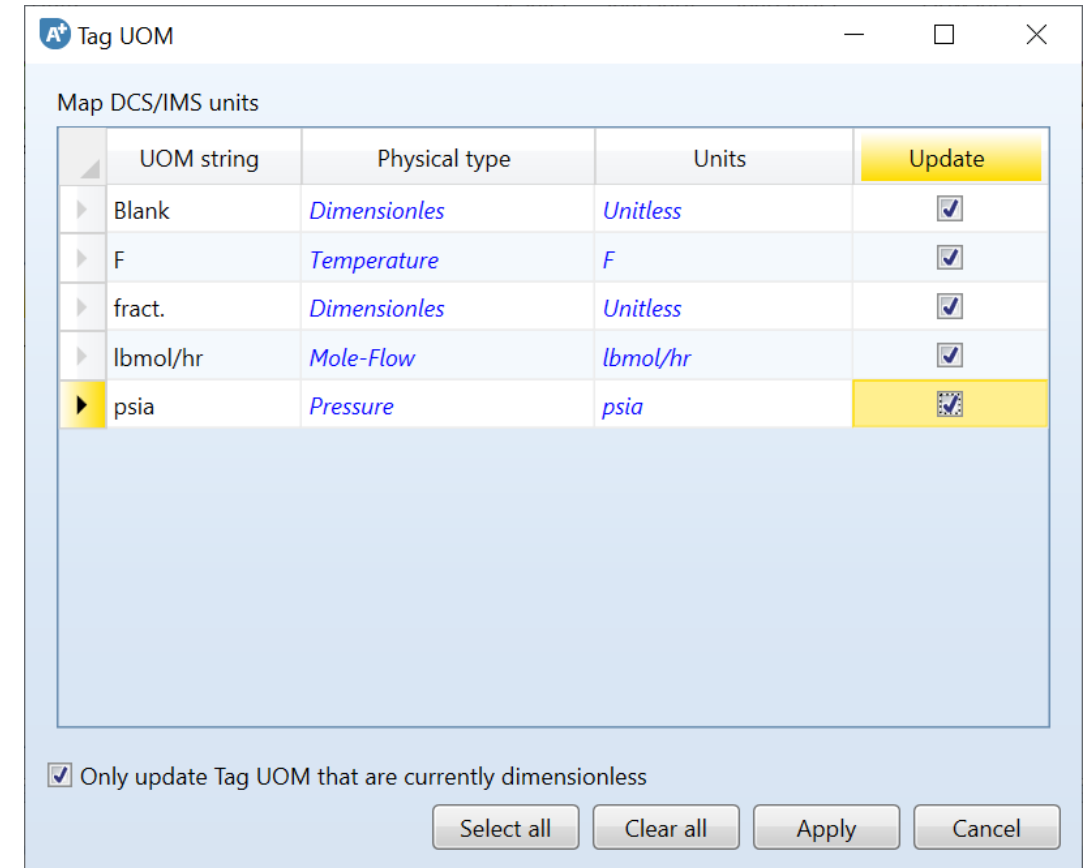
Get Plant Data

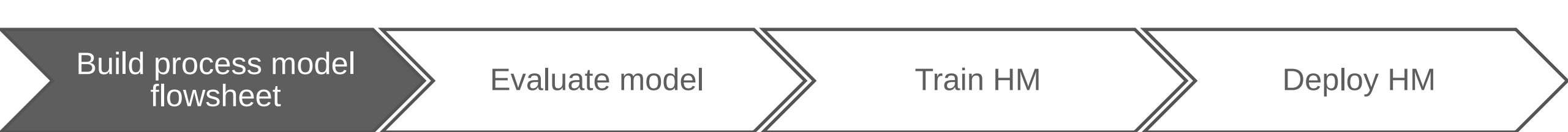
1. Open MSR1.apwz
2. On the Home ribbon, click the AI Training button. Name it as EVALUATE and click OK.
3. On the AI Training ribbon, click Get Plant Data | Import from Excel and select MSR_PlantData.xlsm.
4. Click >> to add all the available datasets and click Import. Once the data successfully imported, close the window.



Tag UOM (Unit of measurement) – step to match physical type and data units with simulation. Mapping the correct physical property is essential.

1. Map the units as shown in the Tag UOM window
2. Select the Update check box and click Apply.
3. Click Done to close the UOM Successfully Updated dialog box.





Build process model
flowsheet

Evaluate model

Train HM

Deploy HM

- Before creating a hybrid model the raw plant data need to be condition.
- There are several conditioning method available.

1. Individual tags

- Data validation
- Steady state detection
- Average

2. Entire dataset

- Bad data slicing
- Pattern search
- Outlier detection
- Smart sampling

Specifications Formula CIM-IO Data Validation Steady-State Detection Initialization Average Manual Input										
Paste from Excel										Details
	Tag name	Units	Tag type	Lower	Upper	Clamp at lower	Clamp at upper	Check frozen	If not good	
▶	001:A1005.PV	F	DCS	1250	1800	☐	☐	☐	Override on bad	
▶	001:A1006.PV	psia	DCS	300	530	☐	☐	☐	Override on bad	
▶	001:A1007.PV	lbmol/hr	DCS	50	450	☐	☐	☐	Override on bad	
▶	001:A1008.PV	Unitless	DCS	0	1	☐	☐	☐	Override on bad	
▶	001:A1009.PV	Unitless	DCS	0	1	☐	☐	☐	Override on bad	

Data validation.

1. On the AI Training ribbon click Configure Conditioning and click Configure Conditioning | Data Validation.
2. Use this sheet to specify data validation parameters for all tags including,
 - Lower and upper limit of a tag
 - Check for frozen values
 - Select clamp to set the tag value to the limiting value if it is outside the corresponding limit.

Configure Conditioning × Main Flowsheet × Control Panel × No Content ×

Control Panel General Bad Data Slicing Pattern Search Outlier Detection Sn

1 Configure Data Conditioning rules
To build a Hybrid Model, at least one conditioned dataset needs to be created

1.a Edit tag specifications...
Edit tags name, description. Add or delete tags, etc.

1.b Conditioning rules for tags
All methods are applied during conditioning. If no conditioning is required, lea

	Conditioning Method	Edit	Configuration form inputs
▶	Data validation	↗	Default Values
▶	Steady-state detection	↗	Default Values
▶		↗	Default Values

Build process model
flowsheet

Evaluate model

Train HM

Deploy HM

1. Establish upper and lower bounds as follows.
 - Reactor T: 1250 – 1800
 - Reactor P: 300 – 530
 - CH₄ mole flow: 50 – 450
 - CO, H₂, CH₄, H₂O mole frac.: 0 -1
2. Click Condition Dataset and click the RUN button. This will process the plant according to the
3. Once complete click Done and Dataset DCR-1 has been created.

A+ Data Conditioning Run

Name:

Description:

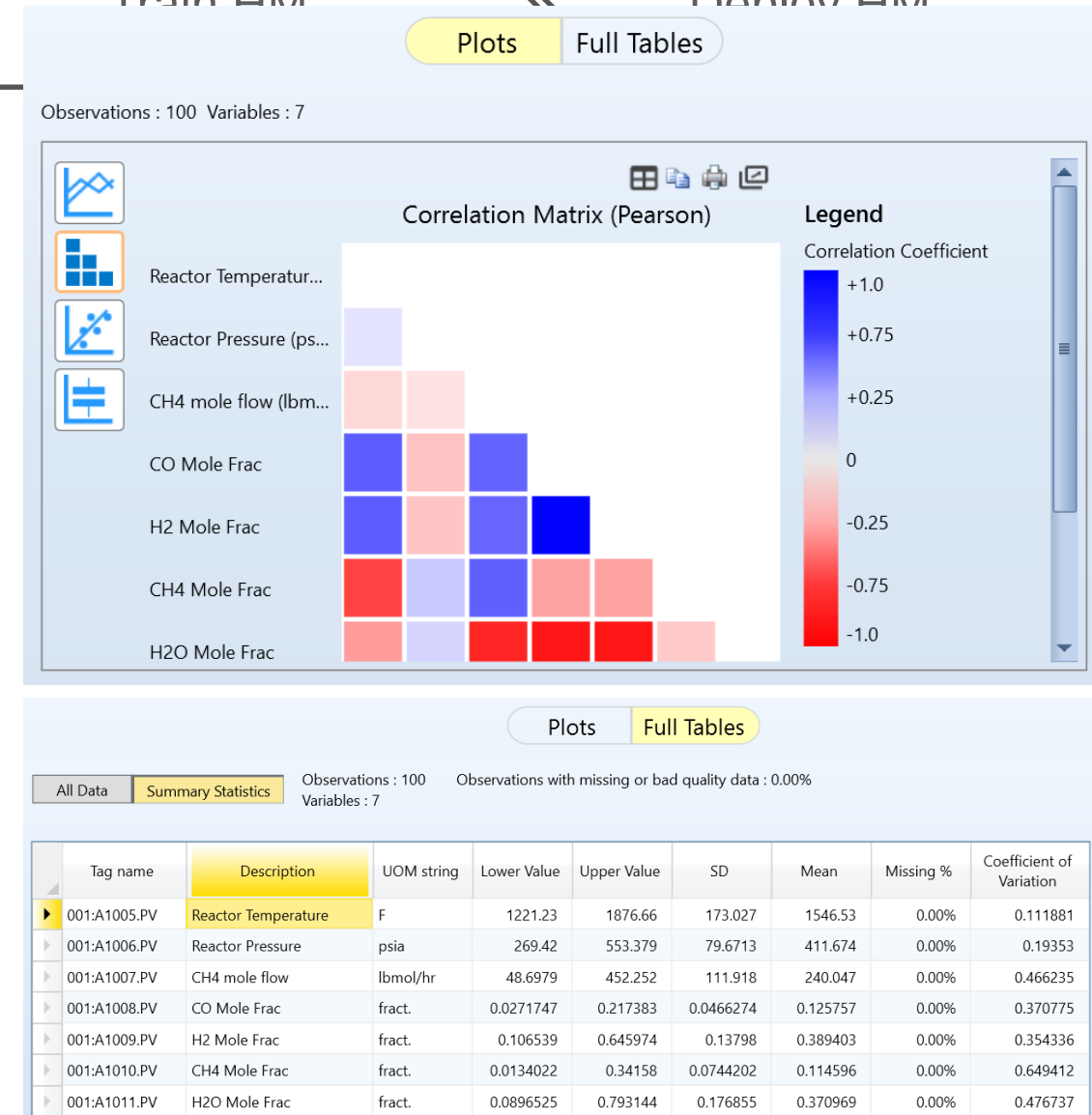
Starting time:

Ending time:

Interval: Days hrs mins

Analyze data.

- There are two forms for data analysis namely Analyze Raw Data and Analyze Conditioned Data.
- This form features two views; Plots View and Full Tables View.
- Plot view has 4 plot options;
 - Trend
 - Correlation matrix
 - Scatter
 - Distribution



Build process model
flowsheet

Evaluate model

Train HM

Deploy HM

All Data

Summary Statistics

Observations : 100

Observations with missing or bad quality data : 0.00%

Variables : 7

	Tag name	Description	UOM string	Lower Value	Upper Value	SD	Mean	Missing %	Coefficient of Variation
▶	001:A1005.PV	Reactor Temperature	F	1221.23	1876.66	173.027	1546.53	0.00%	0.111881
▶	001:A1006.PV	Reactor Pressure	psia	269.42	553.379	79.6713	411.674	0.00%	0.19353
▶	001:A1007.PV	CH4 mole flow	lbmol/hr	48.6979	452.252	111.918	240.047	0.00%	0.466235
▶	001:A1008.PV	CO Mole Frac	fract.	0.0271747	0.217383	0.0466274	0.125757	0.00%	0.370775
▶	001:A1009.PV	H2 Mole Frac	fract.	0.106539	0.645974	0.13798	0.389403	0.00%	0.354336
▶	001:A1010.PV	CH4 Mole Frac	fract.	0.0134022	0.34158	0.0744202	0.114596	0.00%	0.649412
▶	001:A1011.PV	H2O Mole Frac	fract.	0.0896525	0.793144	0.176855	0.370969	0.00%	0.476737

Raw Data Analysis

Dataset

DCR-1

Observations : 74

Variables : 7

All Data

Summary Statistics

	Tag name	Description	Units	Lower Value	Upper Value	SD	Mean	Coefficient of Variation
▶	001:A1005.PV	Reactor Temperature	F	1266.82	1777.36	153.957	1527.07	0.100819
▶	001:A1006.PV	Reactor Pressure	psia	301.056	529.442	66.1938	404.674	0.163573
▶	001:A1007.PV	CH4 mole flow	lbmol/hr	54.6731	432.755	104.241	237.996	0.437997
▶	001:A1008.PV	CO Mole Frac		0.0379813	0.217383	0.0436613	0.12527	0.348539
▶	001:A1009.PV	H2 Mole Frac		0.126848	0.645974	0.125694	0.385662	0.325917
▶	001:A1010.PV	CH4 Mole Frac		0.0196146	0.315286	0.0679634	0.117257	0.579612
▶	001:A1011.PV	H2O Mole Frac		0.105027	0.761924	0.164534	0.372366	0.441861

Conditioned Data Analysis
(Validation method)

Build process model
flowsheet

Evaluate model

Train HM

Deploy HM

Evaluate process model. In this step using specified value of trained variable(s) to calculate the dependent variables at each point in the selected dataset using specified values (initial values) for the neural network outputs (trained parameters).

1. From the AI Training ribbon, click Build Model.
2. Select the mode: Evaluation using specified value of trained variable(s)
3. Select the dataset: DCR-1

Model name: [EVALUATE] Conditioned Dataset: [NA] Snapshot: [NA]

1 Select mode

Mode

- ☐ Training
- ☒ Evaluation using specified value of trained variable(s)
- ☐ Evaluation using calculated value of trained variable(s)

2 Select dataset

Conditioned Dataset DCR-1

Observations : 74

Variables : 7

3 Map simulation input variables as Neural Network Outputs

New

Copy

Paste

Build process model
flowsheet

Evaluate model

Train HM

Deploy HM

3 Map simulation input variables as Neural Network Outputs

New

Copy

Paste

☒ Include in Model	Trained Variable	Description	Lower Limit	Upper Limit	Initial
☒	Reactions.Reactions.MSR.Input.ACT_ENERGY.#0	Lets you type activation energy to...	90000	120000	103224
☒	Reactions.Reactions.MSR.Input.ADS_B.#0.5	Lets you type the coefficient B for...	-10000	-5000	-7500

Map simulation input variables as Neural Network Outputs.

1. Click New to add two variables from the reformer reactor in the Neural Network outputs table.
2. Set the lower and upper limit as well as the initial values.
3. Note: A neural network is not being used here; rather, this just sets up the product windows.

4 Map Tags to simulation variables or to user-defined custom variables
Assign each Tag as Independent or Dependent (simulation inputs = Independent, simulation outputs = Dependent)

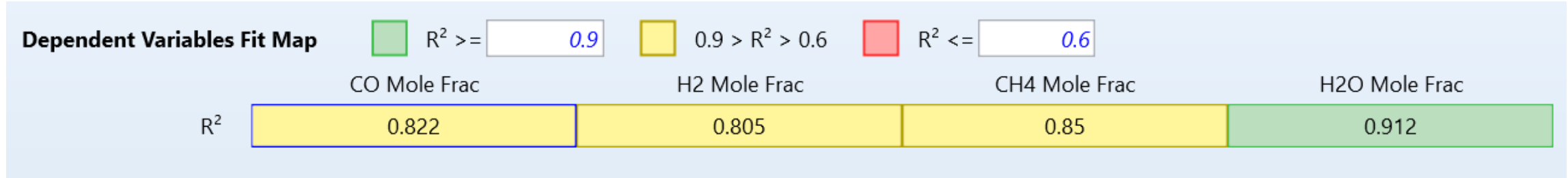
☐ No sorting

Included in training: 7
Excluded : 0
Total : 7

	Tag	Description	Units	Variable Type	Variable Name		Independent	Dependent	
▶	001:A1005.PV	Reactor Temperature	F	Simulation	Blocks.H-01.Input.TEMP	...	☒	☐	
▶	001:A1006.PV	Reactor Pressure	psia	Simulation	Blocks.MIX1.Input.PRES	...	☒	☐	
▶	001:A1007.PV	CH4 mole flow	lbmol/hr	Simulation	Streams.NG.Input.TOTFLOW.MIXED	...	☒	☐	
▶	001:A1008.PV	CO Mole Frac		Simulation	Streams.EFFLUENT.SharedData.Substreams.MIXED.MoleFr	...	☐	☒	
▶	001:A1009.PV	H2 Mole Frac		Simulation	Streams.EFFLUENT.SharedData.Substreams.MIXED.MoleFr	...	☐	☒	
▶	001:A1010.PV	CH4 Mole Frac		Simulation	Streams.EFFLUENT.SharedData.Substreams.MIXED.MoleFr	...	☐	☒	
▶	001:A1011.PV	H2O Mole Frac		Simulation	Streams.EFFLUENT.SharedData.Substreams.MIXED.MoleFr	...	☐	☒	

Map Tags to simulation variables or to user-defined custom variables.

- Add the following simulation variables and select independent (reactor T, reactor P, and CH4 mole flow) or dependent (CO, H2, CH4 and H2O compositions) variables.
- Click Evaluate from the AI Training ribbon.



- There is room to improve the prediction by creating a Hybrid Model that will leverage AI training.
- A Hybrid Model is required to establish a function for the stream EFFULENT
- component mole fractions = $f(\text{independent variables})$, which guarantees a good fit with the available plant data for the dependent variables.

Build process model
flowsheet

Evaluate model

Train HM

Deploy HM

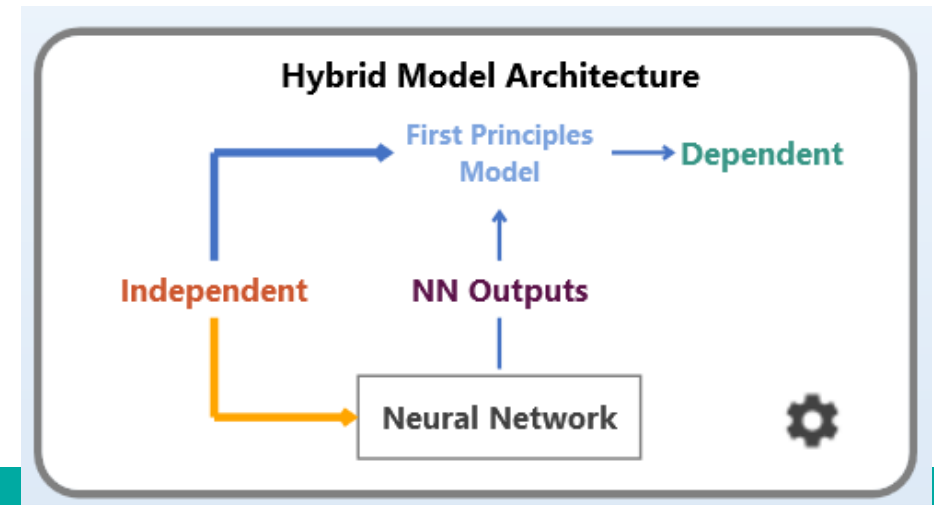
- In the folder AI | AI Training, select Evaluate, and then click Duplicate.
- Rename it as Training.
- In the Navigation Pane, observe the new Training folder. Expand it and select
- Build Model.
- Select the Training mode.
- In the Hybrid Model Architecture open settings.

Model name: [TRAINING] Conditioned Dataset: [DCR-1] Sna

1 Select mode

Mode

- ☒ Training
- ☐ Evaluation using specified value of trained variable(s)
- ☐ Evaluation using calculated value of trained variable(s)



Build process model
flowsheet

Evaluate model

Train HM

Deploy HM

1. Set the Neurons to 16.
2. On the AI Training ribbon click Train to start the training and a training progress windows pops up.
3. When the training is done, close the AI Training Progress plot and observe the results.
4. These settings can be further adjusted to improve the results if needed.

Neural Network Configuration

Basic

Training set fraction: 0.8 Hidden Layers: 1 Training Epochs: 100 Learning Rate: 0.1

Hidden Layer	Neurons	Activation	Activation Parameter
1	16	Selu	0.1

Output Layer

Predicted Parameter	Activation	Activation Parameter	Lower	Upper
Reactions.Reactions.MSR.Input.ACT_ENER...	Sigmoid	0.1	90000	120000
Reactions.Reactions.MSR.Input.ADS_B.#0.5	Sigmoid	0.1	-10000	10000

Initialization

☒ Cold start
☐ Reinitialize from previously converged run

Performance

☒ Use parallel computing during training

Dependent Variables Fit Map

☒ $R^2 \geq 0.9$ ☐ $0.9 > R^2 > 0.6$ ☐ $R^2 \leq 0.6$

CO Mole Frac

H2 Mole Frac

CH4 Mole Frac

H2O Mole Frac

☒ Training ☒ R^2

☐ Validation ☐ R^2

0.949

0.94

0.932

0.978

Build process model
flowsheet

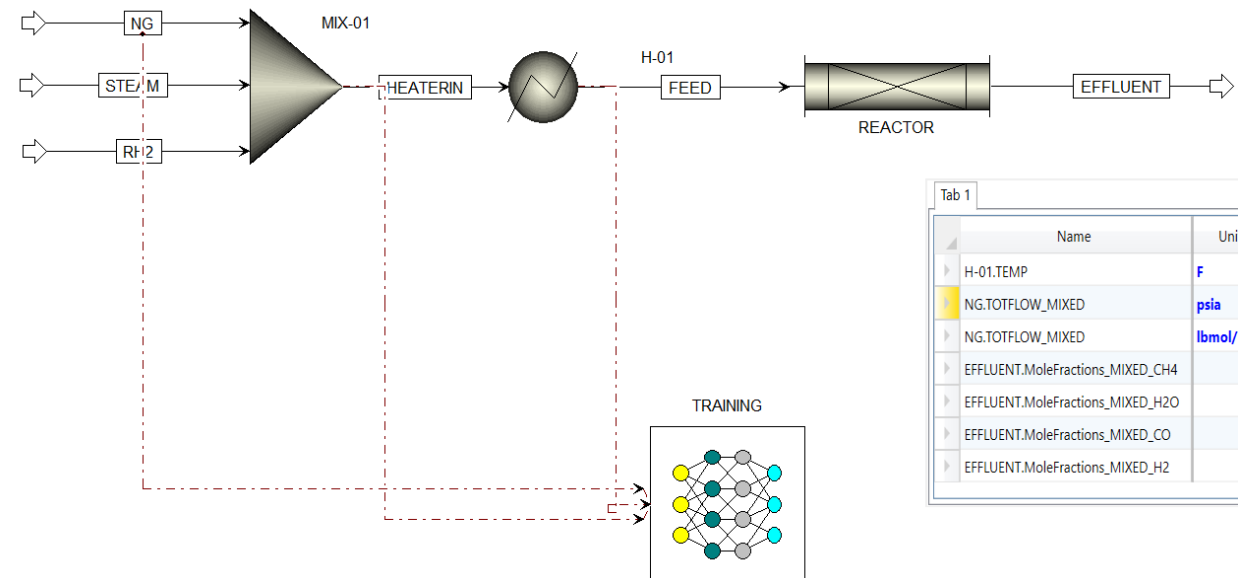
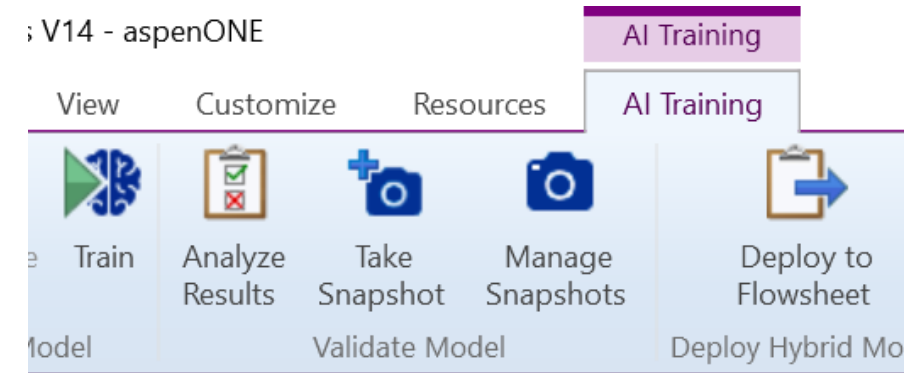
Evaluate model

Train HM

Deploy HM

1. Click Deploy to Flowsheet.
2. A new unit operation with the hybrid model neural network is embedded in the simulation and connected to the reactor.

V14 - aspenONE



Tab 1		
Name	Units	Value
H-01.TEMP	F	1292
NG.TOTFLOW_MIXED	psia	426.183
NG.TOTFLOW_MIXED	lbmol/hr	220.462
EFFLUENT.MoleFractions_MIXED_CH4		0.196035
EFFLUENT.MoleFractions_MIXED_H2O		0.47224
EFFLUENT.MoleFractions_MIXED_CO		0.0801691
EFFLUENT.MoleFractions_MIXED_H2		0.251556